ASSESSMENT

HCL ID : HCLTFP2170249

NAME : DHARUN P

USN : 23BTRCL014

COLLEGE NAME : JAIN UNIVERSITY (FET)

BATCH : 2023-27

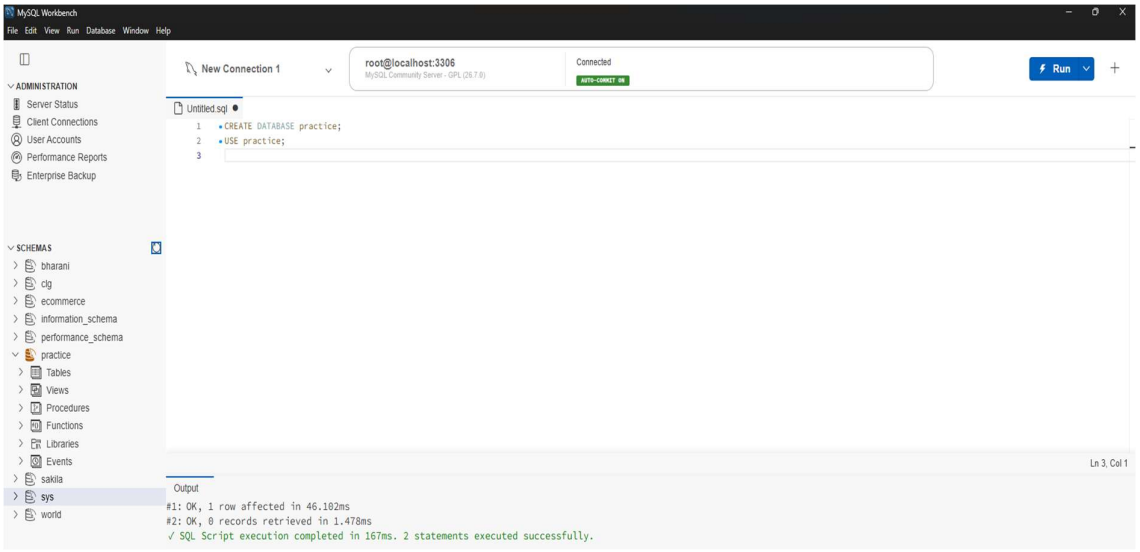
TITLE : AWS CLOUD

&

DATA ENGINEERING

SQL Practice

CREATING A DATABASE AND USING IT.



CREATING TABLES

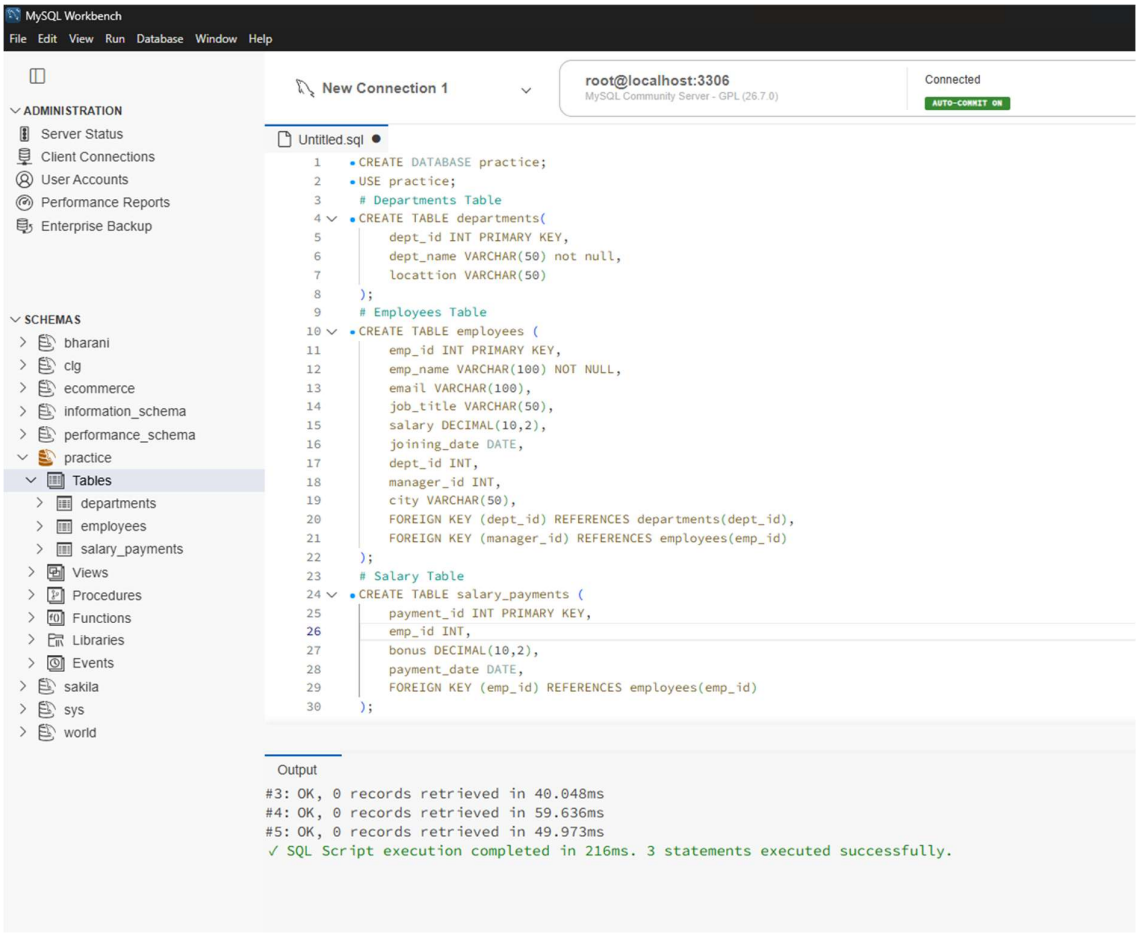


TABLE: DEPARTMENTS

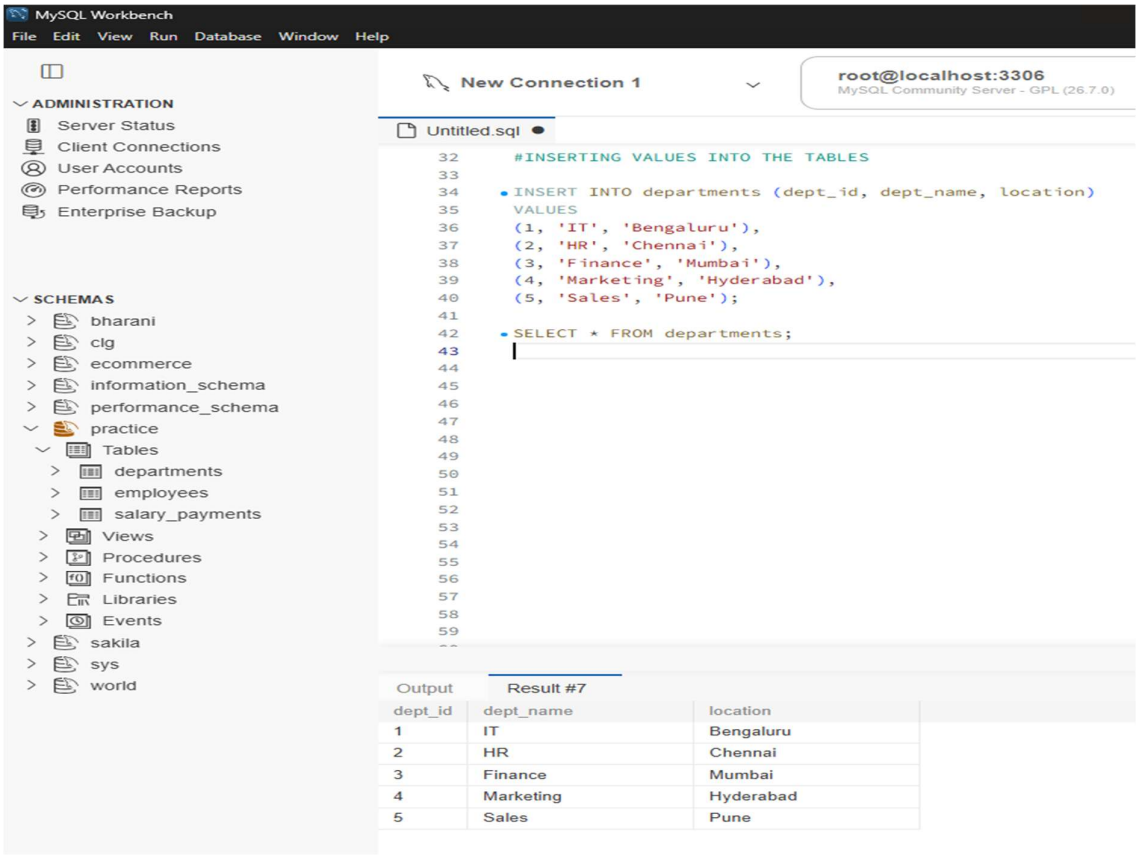


TABLE : EMPLOYEES

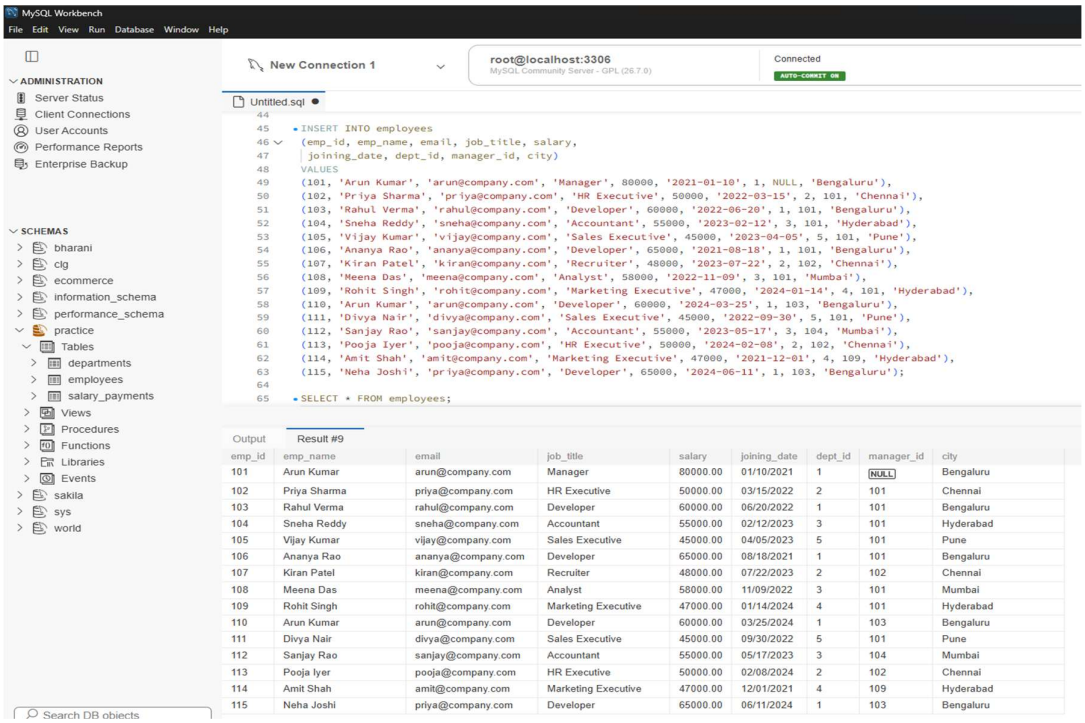
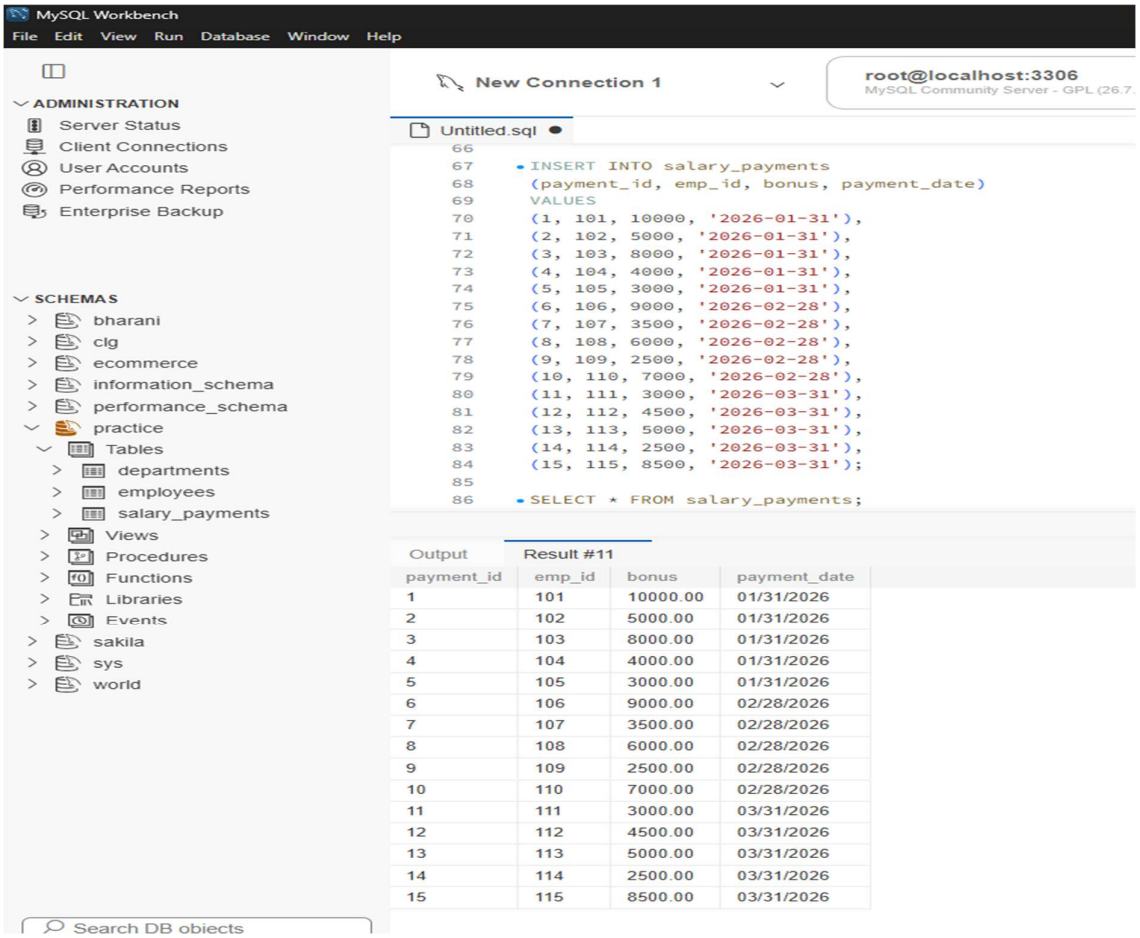
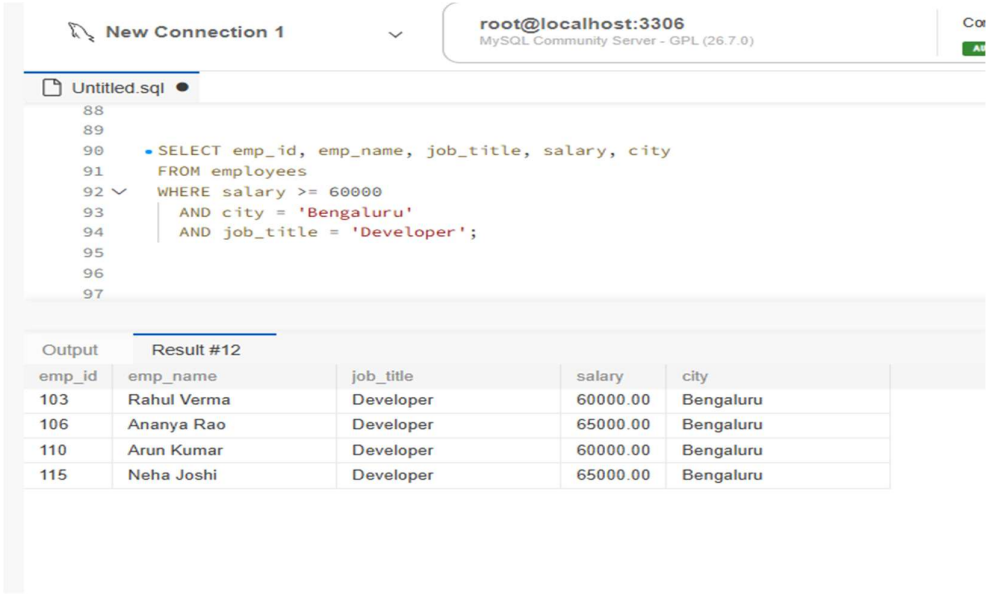


TABLE : SALARY_PAYMENTS



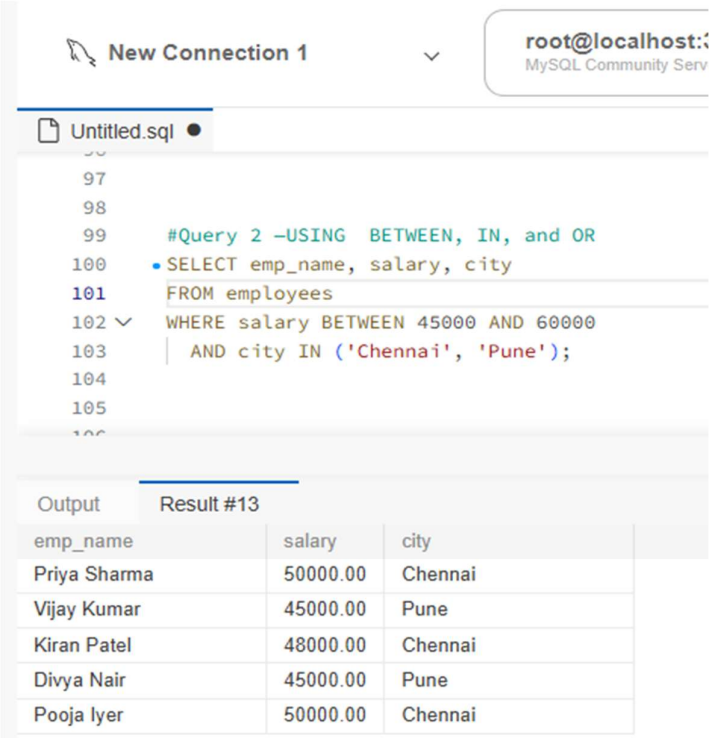
payment_id	emp_id	bonus	payment_date
1	101	10000.00	01/31/2026
2	102	5000.00	01/31/2026
3	103	8000.00	01/31/2026
4	104	4000.00	01/31/2026
5	105	3000.00	01/31/2026
6	106	9000.00	02/28/2026
7	107	3500.00	02/28/2026
8	108	6000.00	02/28/2026
9	109	2500.00	02/28/2026
10	110	7000.00	02/28/2026
11	111	3000.00	03/31/2026
12	112	4500.00	03/31/2026
13	113	5000.00	03/31/2026
14	114	2500.00	03/31/2026
15	115	8500.00	03/31/2026

Query 1 — USING WHERE with multiple conditions



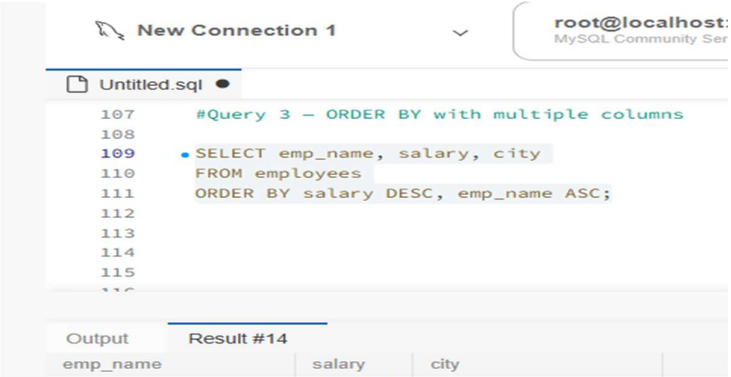
emp_id	emp_name	job_title	salary	city
103	Rahul Verma	Developer	60000.00	Bengaluru
106	Ananya Rao	Developer	65000.00	Bengaluru
110	Arun Kumar	Developer	60000.00	Bengaluru
115	Neha Joshi	Developer	65000.00	Bengaluru

Query 2 —USING BETWEEN, IN, and OR



emp_name	salary	city
Priya Sharma	50000.00	Chennai
Vijay Kumar	45000.00	Pune
Kiran Patel	48000.00	Chennai
Divya Nair	45000.00	Pune
Pooja Iyer	50000.00	Chennai

Query 3 — USING ORDER BY with multiple columns



emp_name	salary	city
Arun Kumar	80000.00	Bengaluru
Ananya Rao	65000.00	Bengaluru
Neha Joshi	65000.00	Bengaluru
Arun Kumar	60000.00	Bengaluru
Rahul Verma	60000.00	Bengaluru
Meena Das	58000.00	Mumbai
Sanjay Rao	55000.00	Mumbai
Sneha Reddy	55000.00	Hyderabad
Pooja Iyer	50000.00	Chennai
Priya Sharma	50000.00	Chennai
Kiran Patel	48000.00	Chennai
Amit Shah	47000.00	Hyderabad
Rohit Singh	47000.00	Hyderabad
Divya Nair	45000.00	Pune
Vijay Kumar	45000.00	Pune

Query 4 — USING GROUP BY with aggregate functions

New Connection 1 — root@localhost:3306 — MySQL Community Server - GPL (26.7.0)

Untitled.sql

```
112
113     #Query 4 – GROUP BY with aggregate functions
114
115     SELECT
116         d.dept_name,
117         COUNT(e.emp_id) AS total_employees,
118         SUM(e.salary) AS total_salary,
119         ROUND(AVG(e.salary), 2) AS average_salary
120     FROM departments d
121     JOIN employees e
122         ON d.dept_id = e.dept_id
123     GROUP BY d.dept_id, d.dept_name;
124
```

Output | Result #15

dept_name	total_employees	total_salary	average_salary
IT	5	330000.00	66000.00
HR	3	148000.00	49333.33
Finance	3	168000.00	56000.00
Marketing	2	94000.00	47000.00
Sales	2	90000.00	45000.00

Query 5 —USING HAVING with GROUP BY

New Connection 1 — root@localhost — MySQL Community Ser

Untitled.sql

```
125     #Query 5 – HAVING with GROUP BY
126
127     SELECT
128         d.dept_name,
129         COUNT(e.emp_id) AS employee_count,
130         AVG(e.salary) AS avg_salary
131     FROM departments d
132     JOIN employees e
133         ON d.dept_id = e.dept_id
134     GROUP BY d.dept_id, d.dept_name
135     HAVING COUNT(e.emp_id) >= 3
136        AND AVG(e.salary) > 50000;
137
```

Output | Result #16

dept_name	employee_count	avg_salary
IT	5	66000.000000
Finance	3	56000.000000

Query 6 — USING INNER JOIN across three tables

Untitled.sql

```
#Query 6 – INNER JOIN across three tables

SELECT
    e.emp_name,
    d.dept_name,
    e.salary,
    s.bonus,
    e.salary + s.bonus AS salary_plus_bonus
FROM employees e
INNER JOIN departments d
    ON e.dept_id = d.dept_id
INNER JOIN salary_payments s
    ON e.emp_id = s.emp_id;
```

Output | Result #17

emp_name	dept_name	salary	bonus	salary_plus_bonus
Arun Kumar	IT	80000.00	10000.00	90000.00
Rahul Verma	IT	60000.00	8000.00	68000.00
Ananya Rao	IT	65000.00	9000.00	74000.00
Arun Kumar	IT	60000.00	7000.00	67000.00
Neha Joshi	IT	65000.00	8500.00	73500.00
Priya Sharma	HR	50000.00	5000.00	55000.00
Kiran Patel	HR	48000.00	3500.00	51500.00
Pooja Iyer	HR	50000.00	5000.00	55000.00
Sneha Reddy	Finance	55000.00	4000.00	59000.00
Meena Das	Finance	58000.00	6000.00	64000.00
Sanjay Rao	Finance	55000.00	4500.00	59500.00
Rohit Singh	Marketing	47000.00	2500.00	49500.00
Amit Shah	Marketing	47000.00	2500.00	49500.00
Vijay Kumar	Sales	45000.00	3000.00	48000.00
Divya Nair	Sales	45000.00	3000.00	48000.00

Query 7 — USING LEFT JOIN to find missing records

New Connection 1 — root@localhost:3306 — MySQL Community Server - GPL (26.7

Untitled.sql

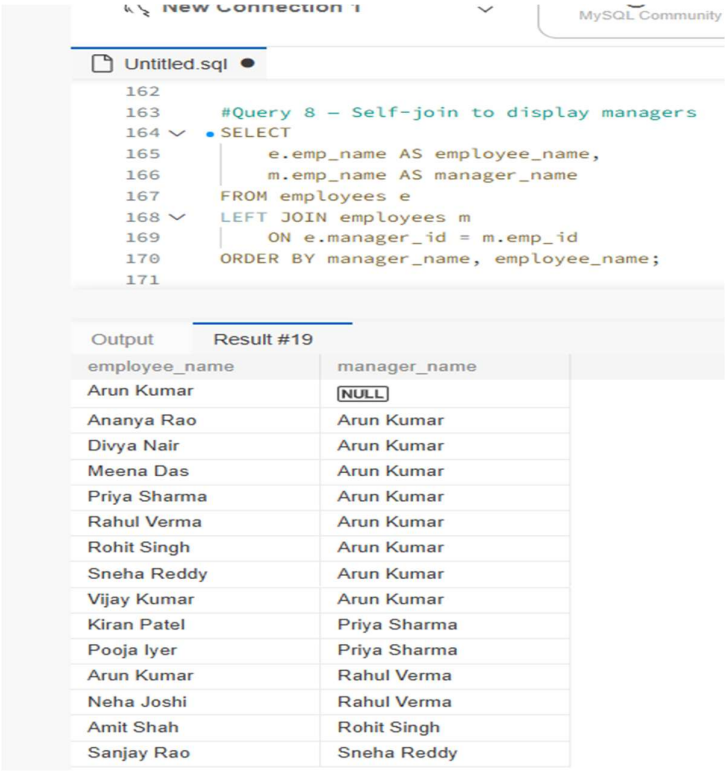
```
#Query 7 – LEFT JOIN to find missing records


SELECT d.dept_id, d.dept_name
FROM departments d
LEFT JOIN employees e
    ON d.dept_id = e.dept_id
WHERE e.emp_id IS NULL;
```

Output | Result #18

*

Query 8 — USING Self-join to display managers

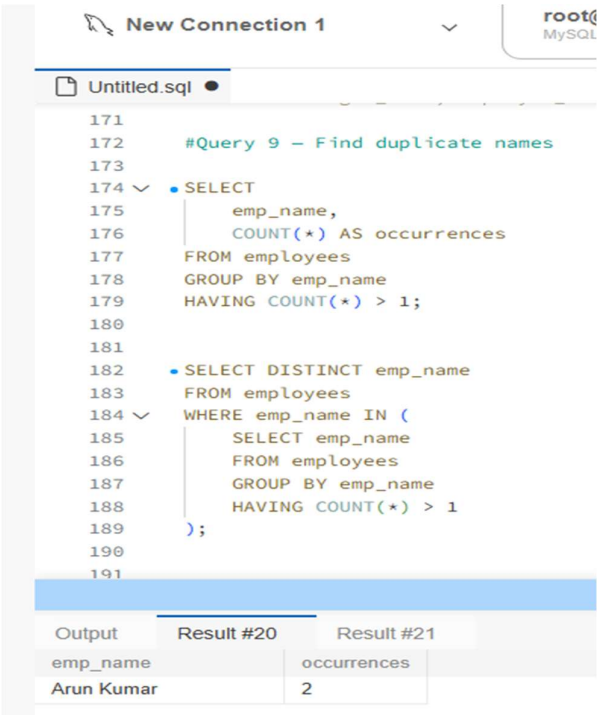


```
#Query 8 – Self-join to display managers
SELECT
    e.emp_name AS employee_name,
    m.emp_name AS manager_name
FROM employees e
LEFT JOIN employees m
    ON e.manager_id = m.emp_id
ORDER BY manager_name, employee_name;
```

Output | Result #19

employee_name	manager_name
Arun Kumar	NULL
Ananya Rao	Arun Kumar
Divya Nair	Arun Kumar
Meena Das	Arun Kumar
Priya Sharma	Arun Kumar
Rahul Verma	Arun Kumar
Rohit Singh	Arun Kumar
Sneha Reddy	Arun Kumar
Vijay Kumar	Arun Kumar
Kiran Patel	Priya Sharma
Pooja Iyer	Priya Sharma
Arun Kumar	Rahul Verma
Neha Joshi	Rahul Verma
Amit Shah	Rohit Singh
Sanjay Rao	Sneha Reddy

Query 9 — Finding duplicate names



```
#Query 9 – Find duplicate names

SELECT
    emp_name,
    COUNT(*) AS occurrences
FROM employees
GROUP BY emp_name
HAVING COUNT(*) > 1;


SELECT DISTINCT emp_name
FROM employees
WHERE emp_name IN (
    SELECT emp_name
    FROM employees
    GROUP BY emp_name
    HAVING COUNT(*) > 1
);
```

Output | Result #20 | Result #21

emp_name	occurrences
Arun Kumar	2

Query 10 — USING Logical functions using IF()

```
191
192     #Query 10 – Logical functions using IF()
193
194
195 ⌄  SELECT
196        emp_name,
197        job_title,
198        salary,
199 ⌄      IF(salary >= 60000, 'Above Threshold', 'Below Threshold')
200            AS salary_status,
201 ⌄      IF(job_title = 'Developer', 'Technical', 'Non-Technical')
202            AS job_category
203    FROM employees;
204
```

Output | Result #22

emp_name	job_title	salary	salary_status	job_category
Arun Kumar	Manager	80000.00	Above Threshold	Non-Technical
Priya Sharma	HR Executive	50000.00	Below Threshold	Non-Technical
Rahul Verma	Developer	60000.00	Above Threshold	Technical
Sneha Reddy	Accountant	55000.00	Below Threshold	Non-Technical
Vijay Kumar	Sales Executive	45000.00	Below Threshold	Non-Technical
Ananya Rao	Developer	65000.00	Above Threshold	Technical
Kiran Patel	Recruiter	48000.00	Below Threshold	Non-Technical
Meena Das	Analyst	58000.00	Below Threshold	Non-Technical
Rohit Singh	Marketing Executive	47000.00	Below Threshold	Non-Technical
Arun Kumar	Developer	60000.00	Above Threshold	Technical
Divya Nair	Sales Executive	45000.00	Below Threshold	Non-Technical
Sanjay Rao	Accountant	55000.00	Below Threshold	Non-Technical
Pooja Iyer	HR Executive	50000.00	Below Threshold	Non-Technical
Amit Shah	Marketing Executive	47000.00	Below Threshold	Non-Technical
Neha Joshi	Developer	65000.00	Above Threshold	Technical

Query 11 —USING Arithmetic operations on salary

New Connection 1 — MySQL Community Server - GPL (26.7.0)

Untitled.sql

```
205     #Query 14 –USING  Arithmetic operations on salary
206
207
208 ⌄  SELECT
209        emp_name,
210        salary AS current_salary,
211        ROUND(salary * 0.10, 2) AS increment_amount,
212        ROUND(salary * 1.10, 2) AS proposed_salary
213    FROM employees
214    ORDER BY proposed_salary DESC;
```

Output | Result #23

emp_name	current_salary	increment_amount	proposed_salary
Arun Kumar	80000.00	8000.00	88000.00
Ananya Rao	65000.00	6500.00	71500.00
Neha Joshi	65000.00	6500.00	71500.00
Rahul Verma	60000.00	6000.00	66000.00
Arun Kumar	60000.00	6000.00	66000.00
Meena Das	58000.00	5800.00	63800.00
Sneha Reddy	55000.00	5500.00	60500.00
Sanjay Rao	55000.00	5500.00	60500.00
Priya Sharma	50000.00	5000.00	55000.00
Pooja Iyer	50000.00	5000.00	55000.00
Kiran Patel	48000.00	4800.00	52800.00
Rohit Singh	47000.00	4700.00	51700.00
Amit Shah	47000.00	4700.00	51700.00
Vijay Kumar	45000.00	4500.00	49500.00
Divya Nair	45000.00	4500.00	49500.00

Query 12 — USING MAX(), MIN(), and AVG()

Untitled.sql

```
#Query 12– MAX(), MIN(), and AVG()


SELECT
    MAX(salary) AS highest_salary,
    MIN(salary) AS lowest_salary,
    AVG(salary) AS average_salary,
    SUM(salary) AS total_monthly_salary,
    COUNT(*) AS employee_count
FROM employees;
```

Output | Result #24

highest_salary	lowest_salary	average_salary	total_monthly_salary	employee_count
80000.00	45000.00	55333.333333	830000.00	15

Query 13 — USING LIKE and pattern matching

Untitled.sql

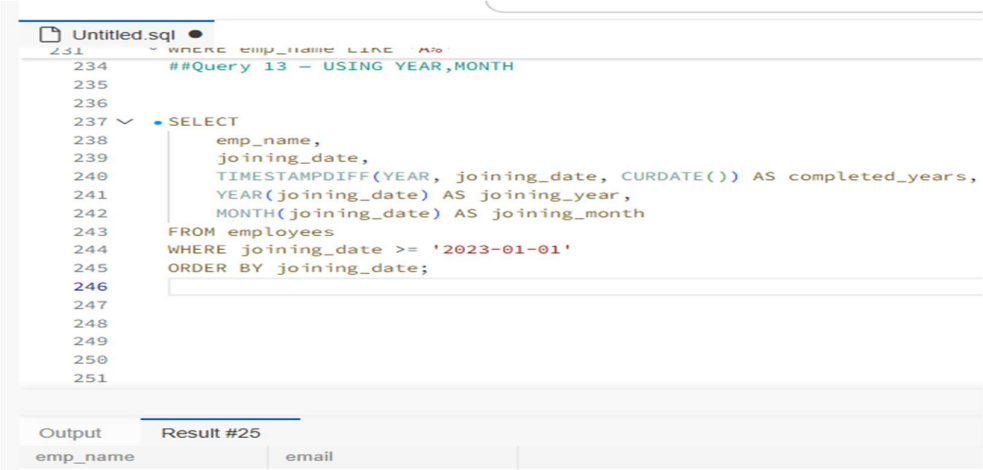
```
#Query 13 – USING LIKE and pattern matching

SELECT emp_name, email
FROM employees
WHERE emp_name LIKE 'A%'
   OR email LIKE '%priya%';
```

Output | Result #25

emp_name	email
Arun Kumar	arun@company.com
Priya Sharma	priya@company.com
Ananya Rao	ananya@company.com
Arun Kumar	arun@company.com
Amit Shah	amit@company.com
Neha Joshi	priya@company.com

#Query 13 — USING YEAR,MONTH



Untitled.sql

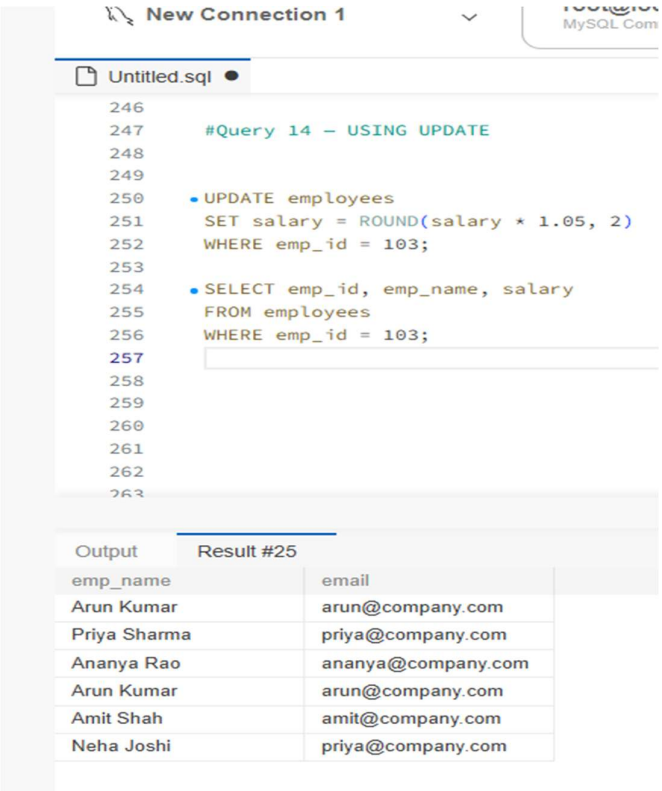
```
##Query 13 — USING YEAR,MONTH

SELECT
    emp_name,
    joining_date,
    TIMESTAMPDIFF(YEAR, joining_date, CURDATE()) AS completed_years,
    YEAR(joining_date) AS joining_year,
    MONTH(joining_date) AS joining_month
FROM employees
WHERE joining_date >= '2023-01-01'
ORDER BY joining_date;
```

Output | Result #25

emp_name	email
Arun Kumar	arun@company.com
Priya Sharma	priya@company.com
Ananya Rao	ananya@company.com
Arun Kumar	arun@company.com
Amit Shah	amit@company.com
Neha Joshi	priya@company.com

Query 14 — USING UPDATE



New Connection 1

Untitled.sql

```
#Query 14 — USING UPDATE

UPDATE employees
SET salary = ROUND(salary * 1.05, 2)
WHERE emp_id = 103;

SELECT emp_id, emp_name, salary
FROM employees
WHERE emp_id = 103;
```

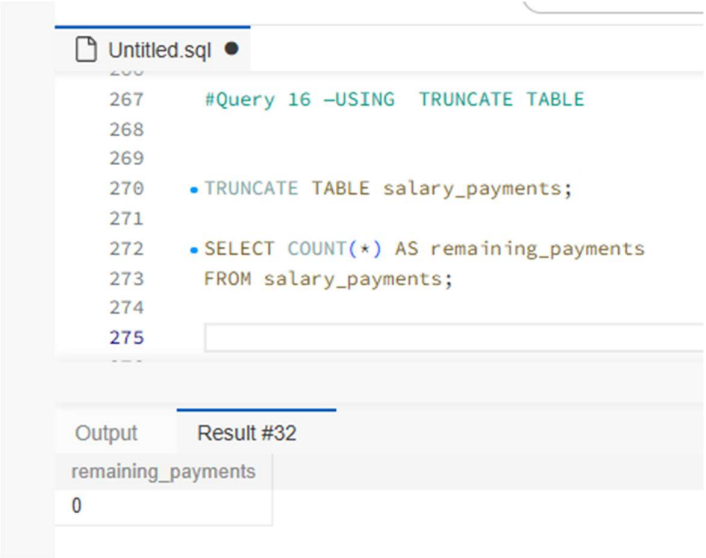
Output | Result #25

emp_name	email
Arun Kumar	arun@company.com
Priya Sharma	priya@company.com
Ananya Rao	ananya@company.com
Arun Kumar	arun@company.com
Amit Shah	amit@company.com
Neha Joshi	priya@company.com

Query 15- USING DELETE with a condition



Query 16- USING TRUNCATE TABLE



Linux Practice



```
Dharun4730@LAPTOP-F2I8Q( X + v

tail: cannot open 'sample_copy.txt' for reading: No such file or directory
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ tail -n 1 renamed.txt
HCL TRAINING
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ wc renamed.txt
 3  8 46 renamed.txt
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ wc -l renamed.txt
3 renamed.txt
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ wc -w renamed.txt
8 renamed.txt
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ wc -c renamed.txt
46 renamed.txt
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ whoami
Dharun4730
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ uname -a
Linux LAPTOP-F2I8QOL7 6.18.40.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Fri Jul 31 22:12:15 UTC 2026 x86_64 GNU/Linux
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ date
Sun Oct 11 13:48:32 UTC 2026
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G     0  3.9G   0% /usr/lib/modules/6.18.40.1-microsoft-standard-WSL2
none            3.9G  4.0K  3.9G   1% /mnt/wsl
drivers         358G  255G  104G  72% /usr/lib/wsl/drivers
/dev/sdd       1007G  1.8G  954G   1% /
none            3.9G   36K  3.9G   1% /mnt/wslg
none            3.9G     0  3.9G   0% /usr/lib/wsl/lib
rootfs          3.9G  3.3M  3.8G   1% /init
none            3.9G  568K  3.9G   1% /run
none            3.9G     0  3.9G   0% /run/lock
none            3.9G     0  3.9G   0% /run/shm
none            3.9G   80K  3.9G   1% /mnt/wslg/versions.txt
none            3.9G   80K  3.9G   1% /mnt/wslg/doc
C:\             358G  255G  104G  72% /mnt/c
D:\             297G  205G   93G  69% /mnt/d
E:\             297G  179G  119G  60% /mnt/e
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           3.9G  8.0K  3.9G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           780M   12K  780M   1% /run/user/1000
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ du -h
0       ./linux_practice
du: cannot read directory './systemd-private-5c1175bcf86a48dda6eb1e6f29b5e4da-systemd-logind.service-0MKus7': Permission denied
0       ./systemd-private-5c1175bcf86a48dda6eb1e6f29b5e4da-systemd-logind.service-0MKus7
du: cannot read directory './systemd-private-5c1175bcf86a48dda6eb1e6f29b5e4da-chrony.service-sTtrA5': Permission denied
0       ./systemd-private-5c1175bcf86a48dda6eb1e6f29b5e4da-chrony.service-sTtrA5
0       ./.X11-unix
du: cannot read directory './snap-private-tmp': Permission denied
0       ./snap-private-tmp
8.0K    .
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G     0  3.9G   0% /usr/lib/modules/6.18.40.1-microsoft-standard-WSL2
none            3.9G  4.0K  3.9G   1% /mnt/wsl
drivers         358G  255G  104G  72% /usr/lib/wsl/drivers
```

```
Dharun4730@LAPTOP-F2I8QOL
tmpfs             780M   12K  780M   1% /run/user/1000
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       654Mi       6.8Gi       4.1Mi       315Mi       7.0Gi
Swap:          2.0Gi          0B       2.0Gi
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ chmod sample.txt
chmod: missing operand
Try 'chmod --help' for more information.
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ sudo whoami
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:

Dharun4730@LAPTOP-F2I8QOL7:/tmp$ sudo apt install nginx
[sudo: authenticate] Password:

Dharun4730@LAPTOP-F2I8QOL7:/tmp$ sudo dnf install nginx
[sudo: authenticate] Password:

Dharun4730@LAPTOP-F2I8QOL7:/tmp$ whoami
Dharun4730
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ sudo whoami
[sudo: authenticate] Password:
root
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [734 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [600 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [173 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [323 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [106 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [508 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [101 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [21.0 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [4168 B]
Get:16 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3136 B]
Get:17 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [138 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [201 kB]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [65.4 kB]
Get:22 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.6 kB]
Get:23 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [495 kB]
Get:24 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [99.6 kB]
Get:25 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [20.0 kB]
Get:26 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [3728 B]
Fetched 4421 kB in 7s (632 kB/s)
103 packages can be upgraded. Run 'apt list --upgradable' to see them.
Dharun4730@LAPTOP-F2I8QOL7:/tmp$
```

```
Dharun4730@LAPTOP-F2I8QOL
103 packages can be upgraded. Run 'apt list --upgradable' to see them.
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ sudo apt install nginx
nginx is already the newest version (1.28.3-2ubuntu1.11).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 103
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ sudo systemctl start nginx
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-10-11 13:31:11 UTC; 26min ago
 Invocation: a5b91c62b9004ef4a91a05b93f5c5ff0
       Docs: man:nginx(8)
    Process: 124 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 191 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 214 (nginx)
      Tasks: 25 (limit: 9340)
     Memory: 18.7M (peak: 22.1M)
        CPU: 77ms
     CGroup: /system.slice/nginx.service
             ├─214 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─216 "nginx: worker process"
             ├─217 "nginx: worker process"
             ├─218 "nginx: worker process"
             ├─219 "nginx: worker process"
             ├─220 "nginx: worker process"
             ├─221 "nginx: worker process"
             ├─222 "nginx: worker process"
             ├─223 "nginx: worker process"
             ├─225 "nginx: worker process"
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             ├─234 "nginx: worker process"
             ├─235 "nginx: worker process"
             ├─236 "nginx: worker process"
             ├─238 "nginx: worker process"
             ├─239 "nginx: worker process"
             ├─240 "nginx: worker process"
             ├─242 "nginx: worker process"
             ├─243 "nginx: worker process"
             ├─244 "nginx: worker process"
             └─245 "nginx: worker process"

Oct 11 13:31:11 LAPTOP-F2I8QOL7 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 11 13:31:11 LAPTOP-F2I8QOL7 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ sudu systemctl enable nginx
Command 'sudu' not found, did you mean:
  command 'tudu' from deb tudu (0.10.4-3)
  command 'sudo' from deb sudo (1.9.17p2-1ubuntu3)
  command 'sudo' from deb sudo-rs (0.2.13-0ubuntu1)
Try: sudo apt install <deb name>
Dharun4730@LAPTOP-F2I8QOL7:/tmp$
```

```
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1480 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:91:80:85 brd ff:ff:ff:ff:ff:ff
    altname enx00155d918085
    inet 172.18.144.2/20 brd 172.18.159.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe91:8085/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ ping -c 4 google.com
PING google.com (142.251.106.139) 56(84) bytes of data.
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=1 ttl=109 time=46.0 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=2 ttl=109 time=45.4 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=3 ttl=109 time=45.8 ms
64 bytes from cm-in-f139.1e100.net (142.251.106.139): icmp_seq=4 ttl=109 time=46.2 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 45.445/45.858/46.180/0.271 ms
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ curl -I https://google.com
HTTP/2 301
location: https://www.google.com/
content-type: text/html; charset=UTF-8
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-k7NS4yNsHtYrkfBRCuxvsA' 'strict-dynamic' 'report
sp.withgoogle.com/csp/gws/other-hp
date: Sun, 11 Oct 2026 14:00:28 GMT
expires: Tue, 10 Nov 2026 14:00:28 GMT
cache-control: public, max-age=2592000
server: gws
content-length: 220
x-xss-protection: 0
x-frame-options: SAMEORIGIN
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

Dharun4730@LAPTOP-F2I8QOL7:/tmp$ cd
Dharun4730@LAPTOP-F2I8QOL7:~$ ssh -i ~/.ssh/Practice.pem ec2-user@YOUR_EC2_PUBLIC_DNS
ssh: Could not resolve hostname your_ec2_public_dns: Name or service not known
Dharun4730@LAPTOP-F2I8QOL7:~$ tar -czf linux_practice.tar.gz notes.txt sample.txt
tar: notes.txt: Cannot stat: No such file or directory
tar: sample.txt: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
Dharun4730@LAPTOP-F2I8QOL7:~$ cd /tmp
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ tar -czf linux_practice.tar.gz notes.txt sample.txt
tar: notes.txt: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
Dharun4730@LAPTOP-F2I8QOL7:/tmp$ history
    1  wsl -l -v
```

AWS

EC2-Elasic Compute Cloud: It is a server that gives access to the user according to their requirement through instances.

IAM- Identity and Access Management: This service helps you control who can access you AWS resources and what actions they are allowed to do.

VPC – Virtual Private cloud: This service helps you to launch resources such as EC2 instances and control how the gateway with the internet and with other resources.

S3- Simple Storage Service: This service is a storage service used to store and retrieve data. Data can be in the form of structured data, unstructured data and also semi-structured data format. It is also know as data lake. Data is stored in buckets.

RDS-Relational Database: This service helps to create, manage, and operate relational databases in the cloud . Data is stored in structured format only (Tables-rows and columns). It is an OLAP tool.

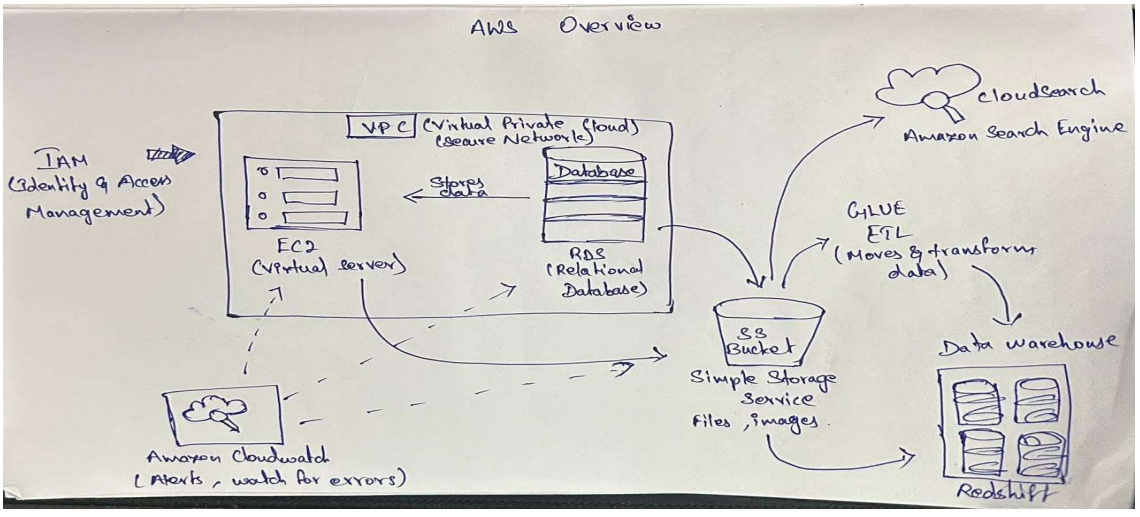
Redshift: This is a service for large data for analytics. It is data warehouse to store and analyze large data using SQL. It stores the data in the form of Structured and semi-structured format.

GLUE: This is service , serverless data integration service used to move and process data from one place to another. It takes raw data , cleans it and transform it and prepare it for analysis. The main components include Glue Crawler, data catalog, database, ETL job, Visual ETL.

GLUE ETL- Extract, Transform, Load. It is the process of taking data from a source , cleaning and loading the processed data into a location.

CloudSearch: It is a Service that is used to search the exact data.

CloudWatch: It is a AWS monitoring service that is used to monitor your AWS resources and application whether the AWS services are working properly.



9/10/2026

Hospital Management System

IAM - Access and Permission

Doctors, nurses, receptionist, and administrators receive role based access.

↓

VPC - Virtual Private Cloud

Hospitals private Network

Public entry point, private databases, private web applications.

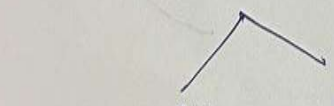
↓

EC2 - Elastic Compute Cloud

Application

→ Create an instance.

→ Instance is nothing but virtual server.



Amazon RDS

Relational Database

Patient profiles, Appointment etc.

Structured

Amazon S3

Scans, reports, Images.

Stored in Buckets.

↓

GLUE (Process the data)

Data Preparation

ETL - process.

cleans the data



Redshift

Large data Analysis

Hospital-wide analysis

Athena

SQL query on S3 Data.

↓

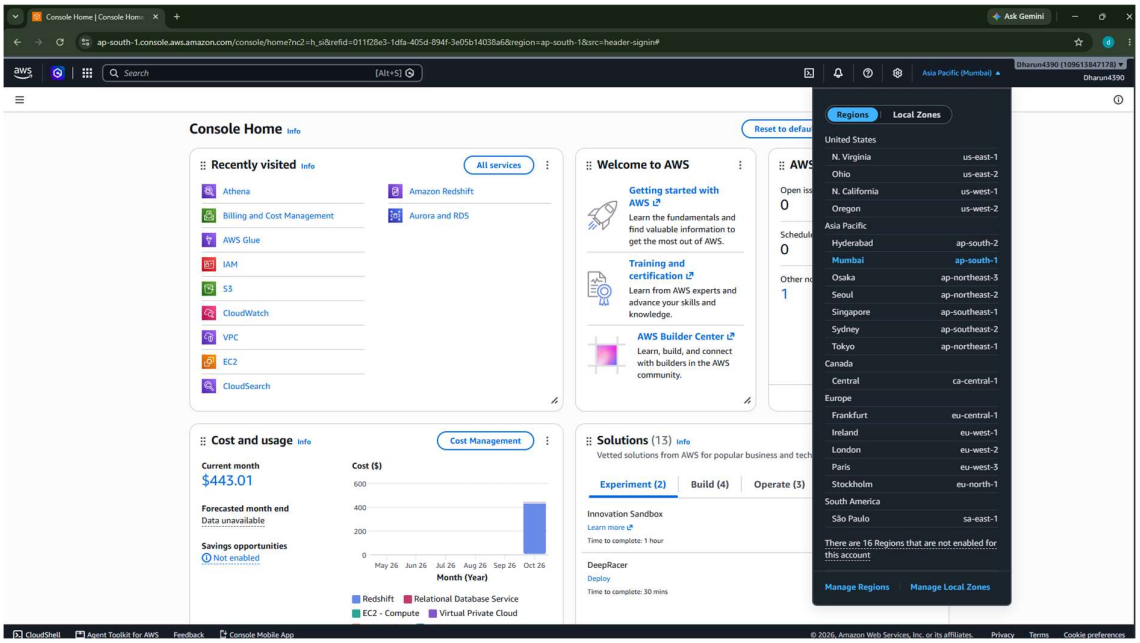
Cloud Search

Search doctors, departments and approved searchable records.

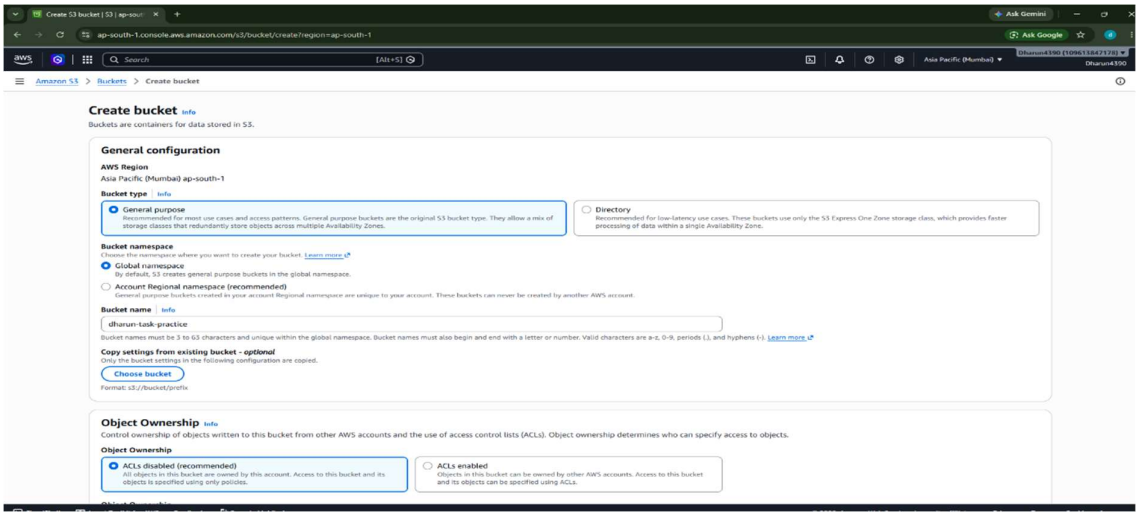
Cloud Watch

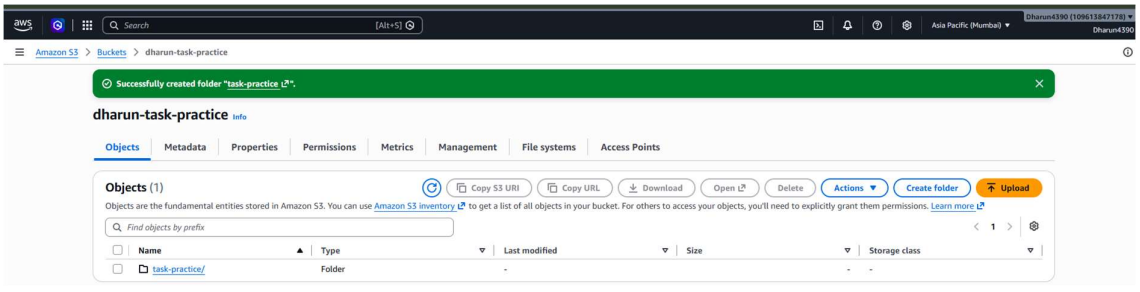
Monitors application errors, database performance, logs and alerts across the system.

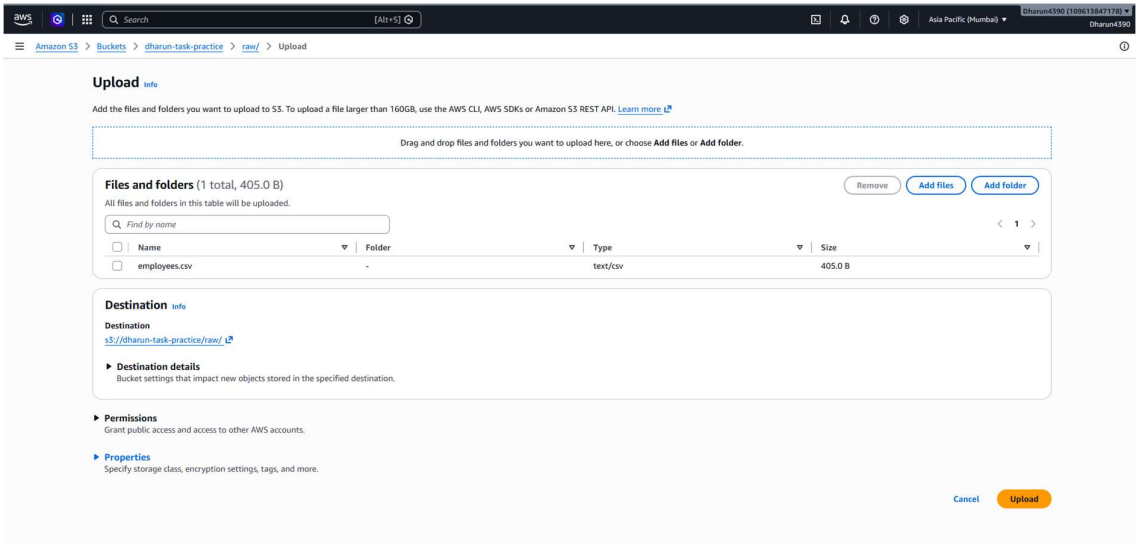
AWS Account / Region

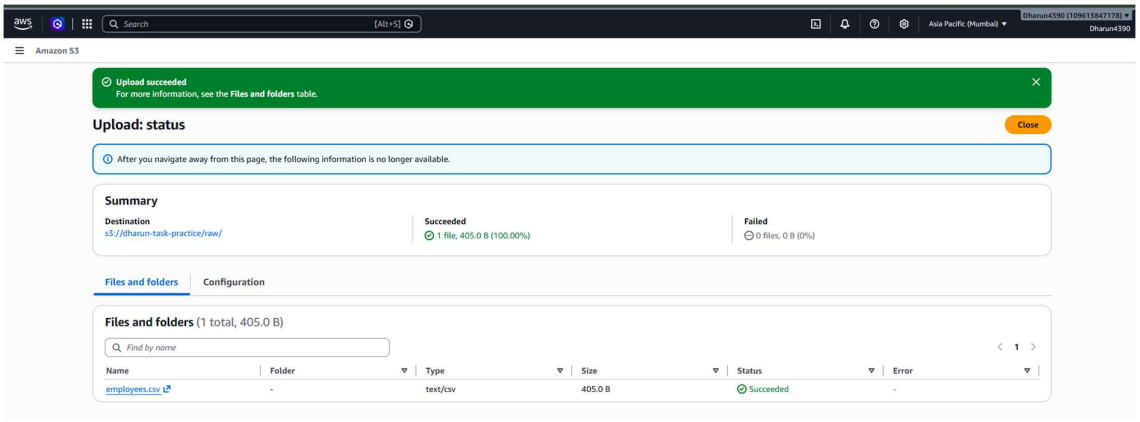


Amazon S3

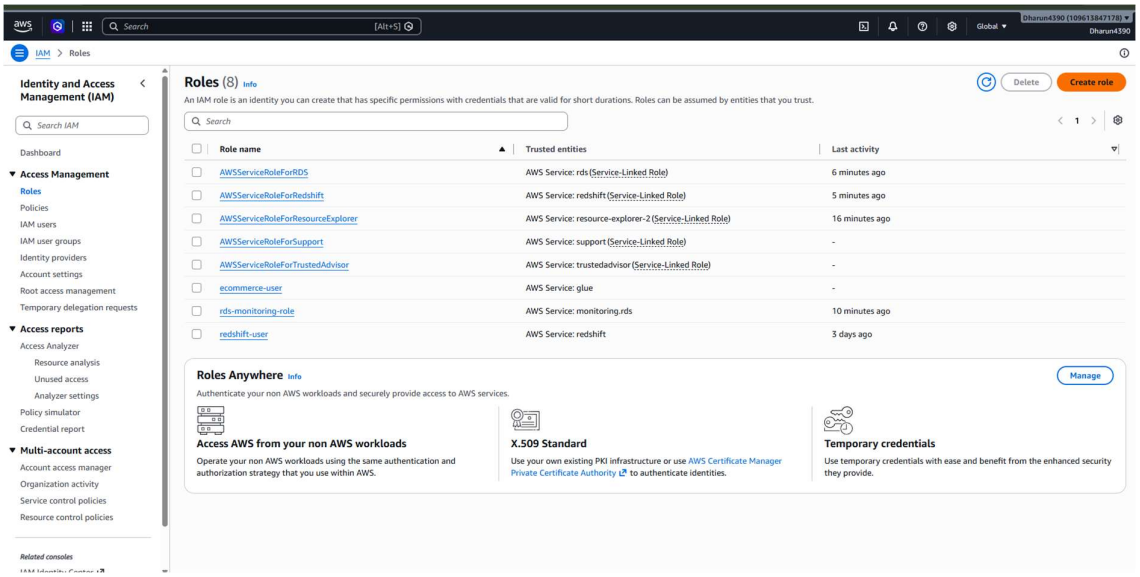


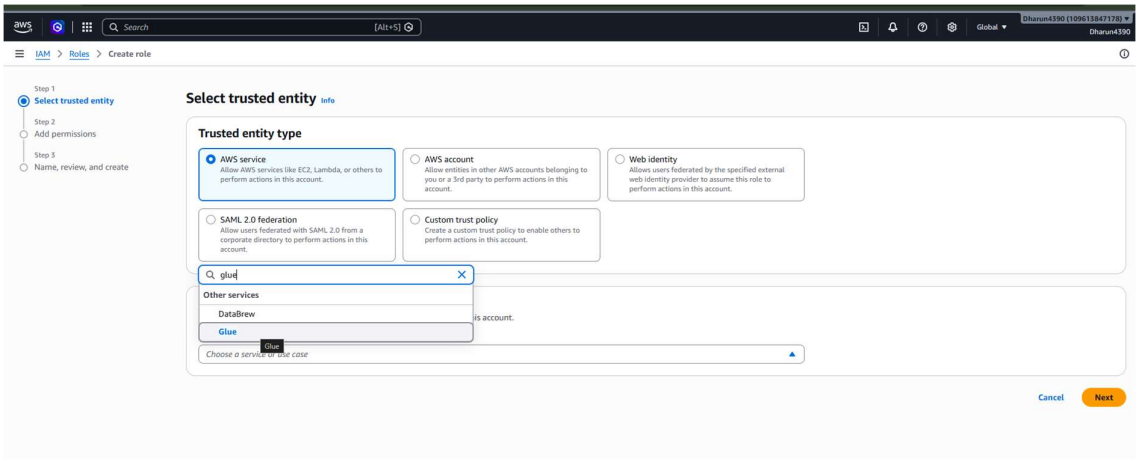


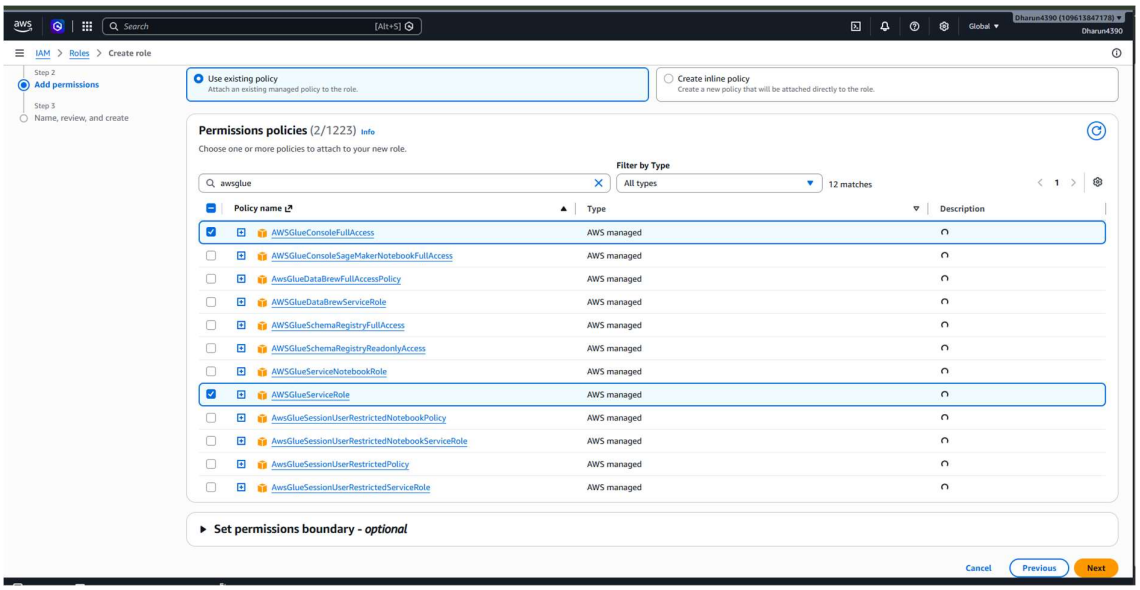

Search [Alt+S]
Asia Pacific (Mumbai)
Dharun4390 (109613847178)
Dharun4390
Amazon S3 > Buckets > dharun-task-practice > raw/ > Upload
Upload Info
Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. Learn more
Drag and drop files and folders you want to upload here, or choose Add files or Add folder.
Files and folders (1 total, 405.0 B)
All files and folders in this table will be uploaded.
Remove
Add files
Add folder
Find by name
Name
Folder
Type
Size
employees.csv
-
text/csv
405.0 B
Destination Info
Destination
s3://dharun-task-practice/raw/
Destination details
Bucket settings that impact new objects stored in the specified destination.
Permissions
Grant public access and access to other AWS accounts.
Properties
Specify storage class, encryption settings, tags, and more.
Cancel
Upload

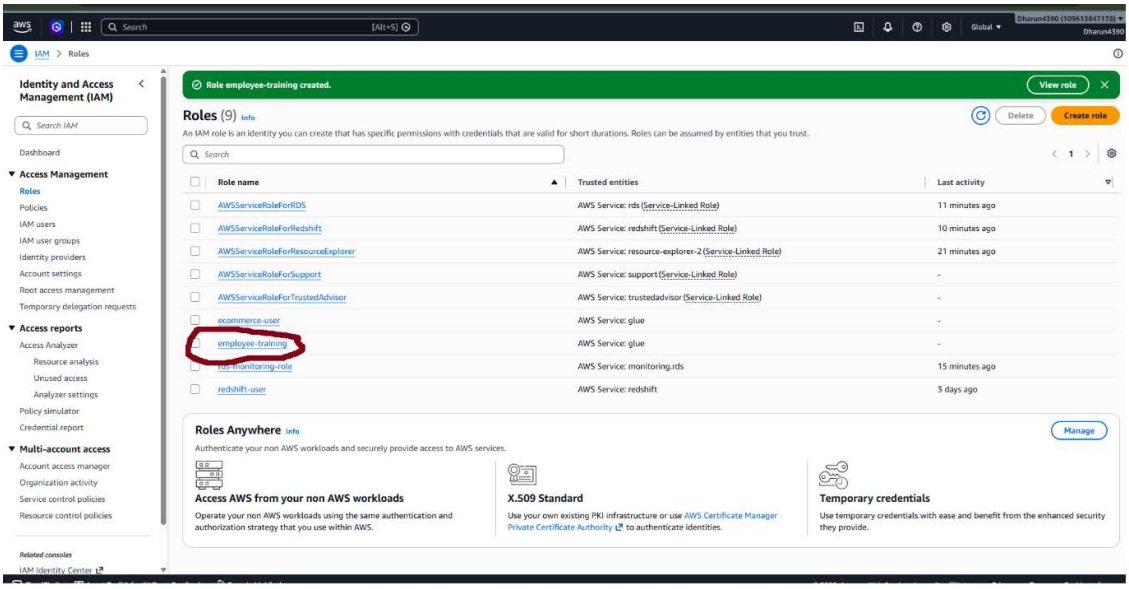

Search [Alt+S]
Asia Pacific (Mumbai)
Dharun4390 (109613847178)
Dharun4390
Amazon S3
Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://dharun-task-practice/raw/
Succeeded
1 file, 405.0 B (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (1 total, 405.0 B)
Find by name
Name
Folder
Type
Size
Status
Error
employees.csv
-
text/csv
405.0 B
Succeeded
-

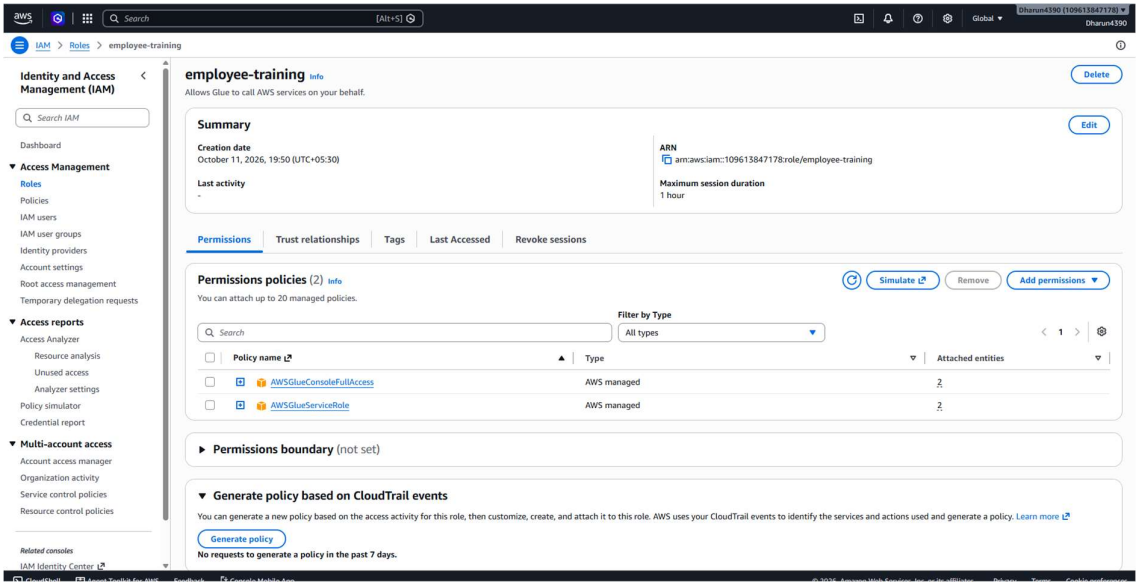
IAM


Search [Alt+S]
Global
Dharun4390 (109613847178)
Dharun4390
IAM > Roles
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Credential report
Multi-account access
Account access manager
Organization activity
Service control policies
Resource control policies
Related consoles
Roles (8) Info
An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.
Delete
Create role
Search
Role name
Trusted entities
Last activity
AWSServiceRoleForRDS
AWS Service: rds (Service-Linked Role)
6 minutes ago
AWSServiceRoleForRedshift
AWS Service: redshift (Service-Linked Role)
5 minutes ago
AWSServiceRoleForResourceExplorer
AWS Service: resource-explorer-2 (Service-Linked Role)
16 minutes ago
AWSServiceRoleForSupport
AWS Service: support (Service-Linked Role)
-
AWSServiceRoleForTrustedAdvisor
AWS Service: trustedadvisor (Service-Linked Role)
-
ecommerce-user
AWS Service: glue
-
rds-monitoring-role
AWS Service: monitoring.rds
10 minutes ago
redshift-user
AWS Service: redshift
3 days ago
Roles Anywhere Info
Manage
Authenticate your non AWS workloads and securely provide access to AWS services.
Access AWS from your non AWS workloads
Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.
X.509 Standard
Use your own existing PKI infrastructure or use AWS Certificate Manager Private Certificate Authority to authenticate identities.
Temporary credentials
Use temporary credentials with ease and benefit from the enhanced security they provide.

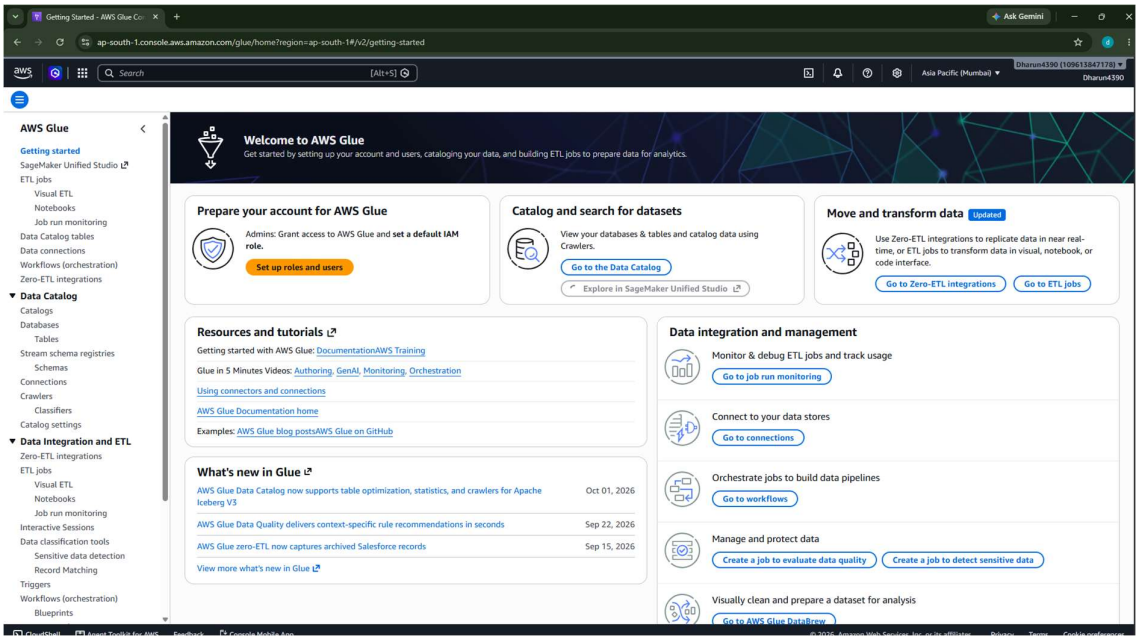

IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Select trusted entity Info
Trusted entity type
AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.
glue
Other services
DataBrew
Glue
Glue
Choose a service or use case
Cancel
Next

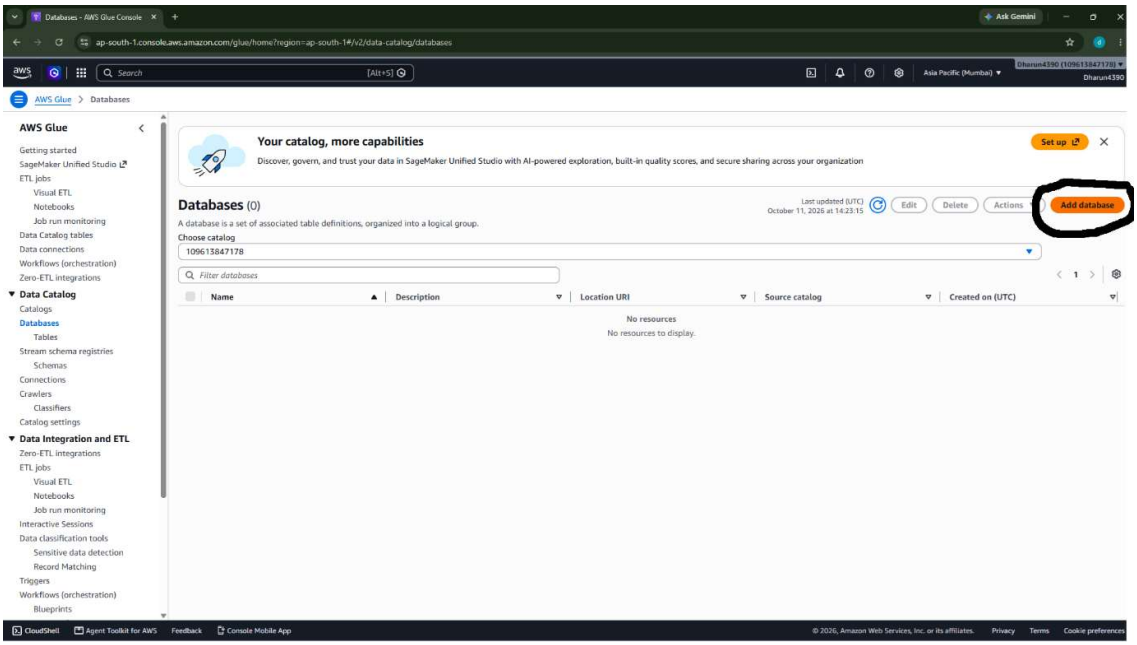

IAM > Roles > Create role
Step 2
Add permissions
Step 3
Name, review, and create
Use existing policy
Attach an existing managed policy to the role.
Create inline policy
Create a new policy that will be attached directly to the role.
Permissions policies (2/1223) Info
Choose one or more policies to attach to your new role.
awsglue
Filter by Type
All types
12 matches
Policy name
Type
Description
AWSGlueConsoleFullAccess AWS managed
AWSGlueConsoleSageMakerNotebookFullAccess AWS managed
AwsGlueDataBrewFullAccessPolicy AWS managed
AWSGlueDataBrewServiceRole AWS managed
AWSGlueSchemaRegistryFullAccess AWS managed
AWSGlueSchemaRegistryReadonlyAccess AWS managed
AWSGlueServiceNotebookRole AWS managed
AWSGlueServiceRole AWS managed
AwsGlueSessionUserRestrictedNotebookPolicy AWS managed
AwsGlueSessionUserRestrictedNotebookServiceRole AWS managed
AwsGlueSessionUserRestrictedPolicy AWS managed
AwsGlueSessionUserRestrictedServiceRole AWS managed
Set permissions boundary - optional
Cancel
Previous
Next

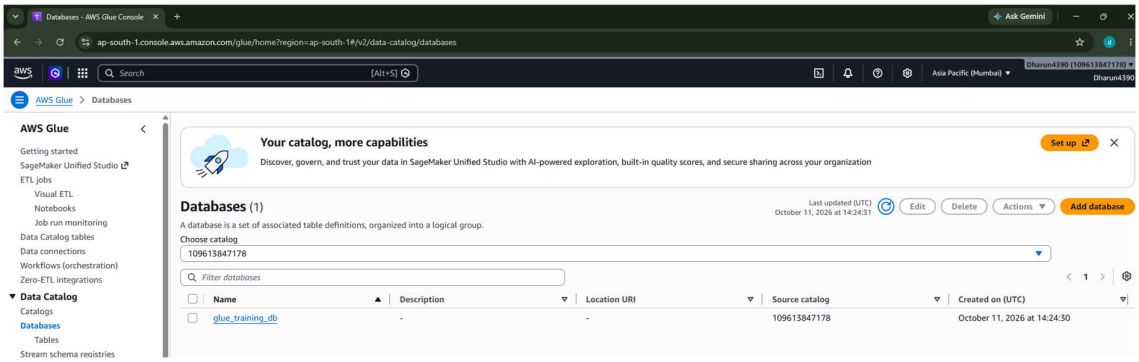

IAM > Roles
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Credential report
Multi-account access
Account access manager
Organization activity
Service control policies
Resource control policies
Related consoles
IAM Identity Center
Role employee-training created.
View role
Roles (9) Info
An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.
Delete
Create role
Search
Role name
Trusted entities
Last activity
AWSServiceRoleForRDS AWS Service: rds (Service-Linked Role) 11 minutes ago
AWSServiceRoleForRedshift AWS Service: redshift (Service-Linked Role) 10 minutes ago
AWSServiceRoleForResourceExplorer AWS Service: resource-explorer-2 (Service-Linked Role) 21 minutes ago
AWSServiceRoleForSupport AWS Service: support (Service-Linked Role) -
AWSServiceRoleForTrustedAdvisor AWS Service: trustedadvisor (Service-Linked Role) -
ecommerce-user AWS Service: glue -
employee-training AWS Service: glue -
rds-monitoring-role AWS Service: monitoring.rds 15 minutes ago
redshift-user AWS Service: redshift 3 days ago
Roles Anywhere Info
Manage
Authenticate your non AWS workloads and securely provide access to AWS services.
Access AWS from your non AWS workloads
Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.
X.509 Standard
Use your own existing PKI infrastructure or use AWS Certificate Manager Private Certificate Authority to authenticate identities.
Temporary credentials
Use temporary credentials with ease and benefit from the enhanced security they provide.

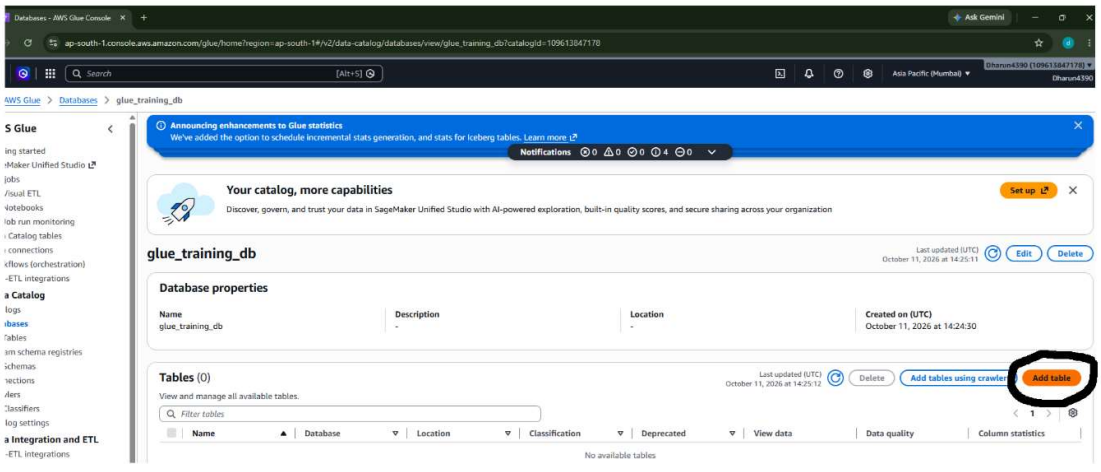


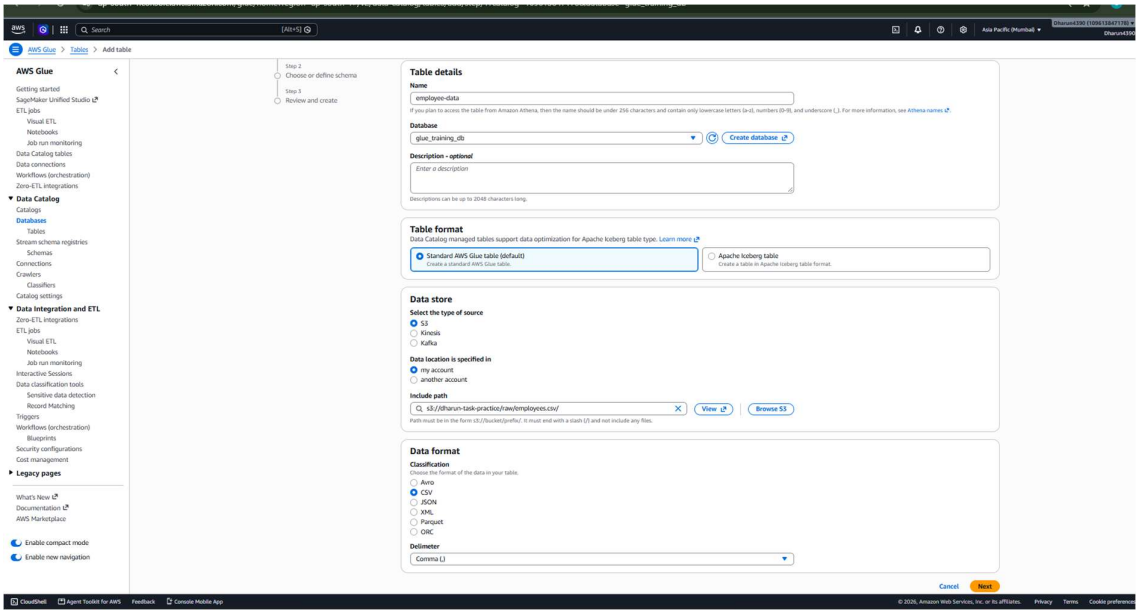
AWS Glue Database

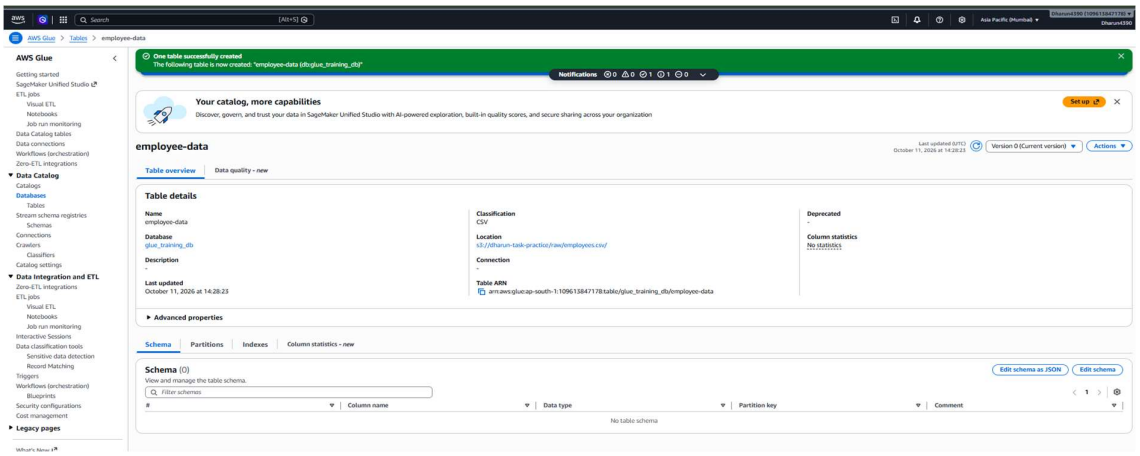



Databases - AWS Glue Console
ap-south-1.console.aws.amazon.com/glue/home?region=ap-south-1#/v2/data-catalog/databases
Ask Gemini
Search
[Alt+S]
Asia Pacific (Mumbai)
Dharun4390
AWS Glue > Databases
AWS Glue
Getting started
SageMaker Unified Studio
ETL jobs
Visual ETL
Notebooks
Job run monitoring
Data Catalog tables
Data connections
Workflows (orchestration)
Zero-ETL integrations
Data Catalog
Catalogs
Databases
Tables
Stream schema registries
Schemas
Connections
Crawlers
Classifiers
Catalog settings
Data Integration and ETL
Zero-ETL integrations
ETL jobs
Visual ETL
Notebooks
Job run monitoring
Interactive Sessions
Data classification tools
Sensitive data detection
Record Matching
Triggers
Workflows (orchestration)
Blueprints
Your catalog, more capabilities
Discover, govern, and trust your data in SageMaker Unified Studio with AI-powered exploration, built-in quality scores, and secure sharing across your organization
Set up
Databases (0)
A database is a set of associated table definitions, organized into a logical group.
Last updated (UTC) October 11, 2026 at 14:23:15
Edit
Delete
Actions
Add database
Choose catalog
109613847178
Filter databases
Name
Description
Location URI
Source catalog
Created on (UTC)
No resources
No resources to display.
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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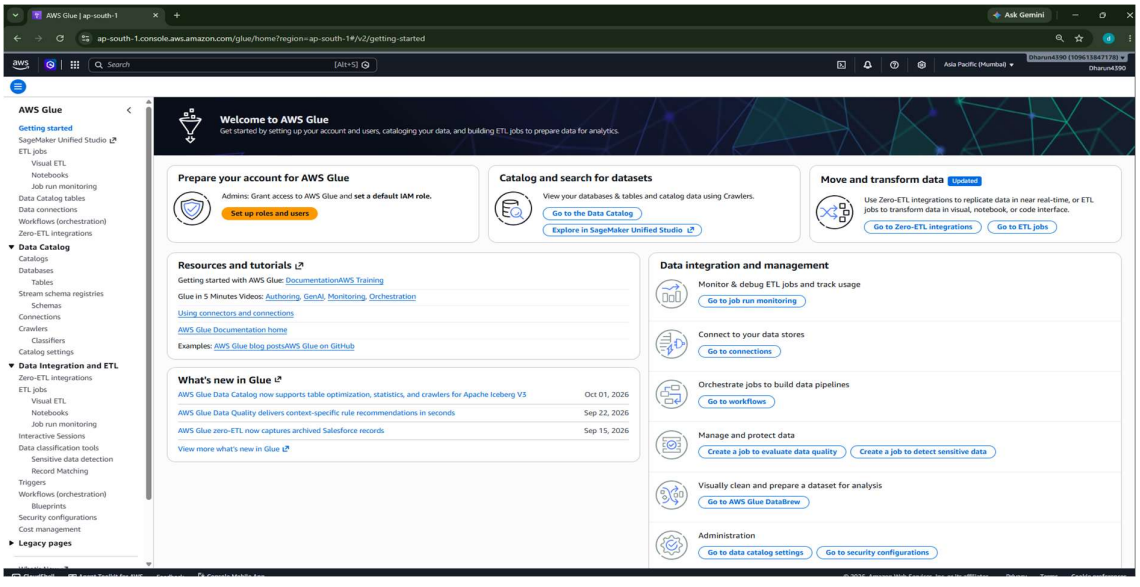

Databases - AWS Glue Console
ap-south-1.console.aws.amazon.com/glue/home?region=ap-south-1#/v2/data-catalog/databases
Ask Gemini
Search
[Alt+S]
Asia Pacific (Mumbai)
Dharun4390
AWS Glue > Databases
AWS Glue
Getting started
SageMaker Unified Studio
ETL jobs
Visual ETL
Notebooks
Job run monitoring
Data Catalog tables
Data connections
Workflows (orchestration)
Zero-ETL integrations
Data Catalog
Catalogs
Databases
Tables
Stream schema registries
Your catalog, more capabilities
Discover, govern, and trust your data in SageMaker Unified Studio with AI-powered exploration, built-in quality scores, and secure sharing across your organization
Set up
Databases (1)
A database is a set of associated table definitions, organized into a logical group.
Last updated (UTC) October 11, 2026 at 14:24:31
Edit
Delete
Actions
Add database
Choose catalog
109613847178
Filter databases
Name
Description
Location URI
Source catalog
Created on (UTC)
glue_training_db
-
-
109613847178
October 11, 2026 at 14:24:30

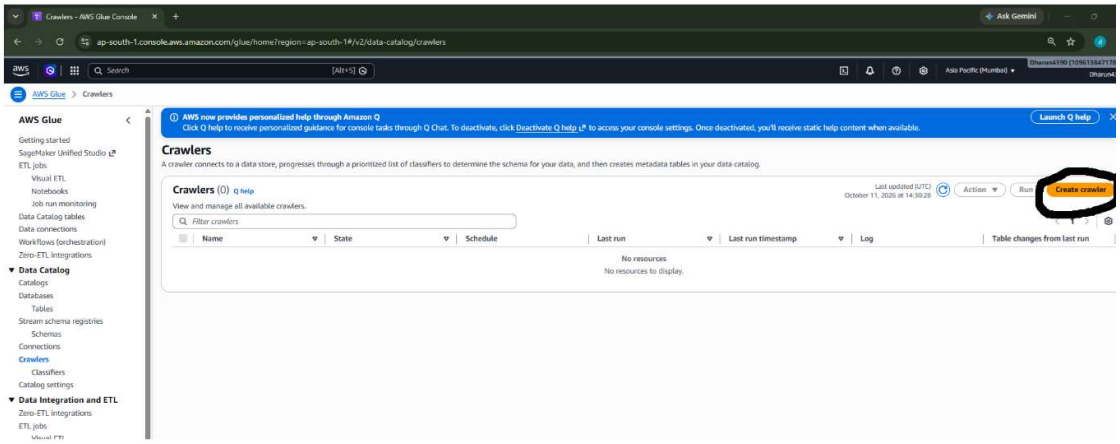

Databases - AWS Glue Console
ap-south-1.console.aws.amazon.com/glue/home?region=ap-south-1#/v2/data-catalog/databases/view/glue_training_db?catalogId=109613847178
Ask Gemini
Search
[Alt+S]
Asia Pacific (Mumbai)
Dharun4390
AWS Glue > Databases > glue_training_db
Announcing enhancements to Glue statistics
We've added the option to schedule incremental stats generation, and stats for iceberg tables. Learn more
Notifications
Your catalog, more capabilities
Discover, govern, and trust your data in SageMaker Unified Studio with AI-powered exploration, built-in quality scores, and secure sharing across your organization
Set up
glue_training_db
Last updated (UTC) October 11, 2026 at 14:25:11
Edit
Delete
Database properties
Name
glue_training_db
Description
-
Location
-
Created on (UTC)
October 11, 2026 at 14:24:30
Tables (0)
View and manage all available tables.
Last updated (UTC) October 11, 2026 at 14:25:12
Delete
Add tables using crawler
Add table
Filter tables
Name
Database
Location
Classification
Deprecated
View data
Data quality
Column statistics
No available tables

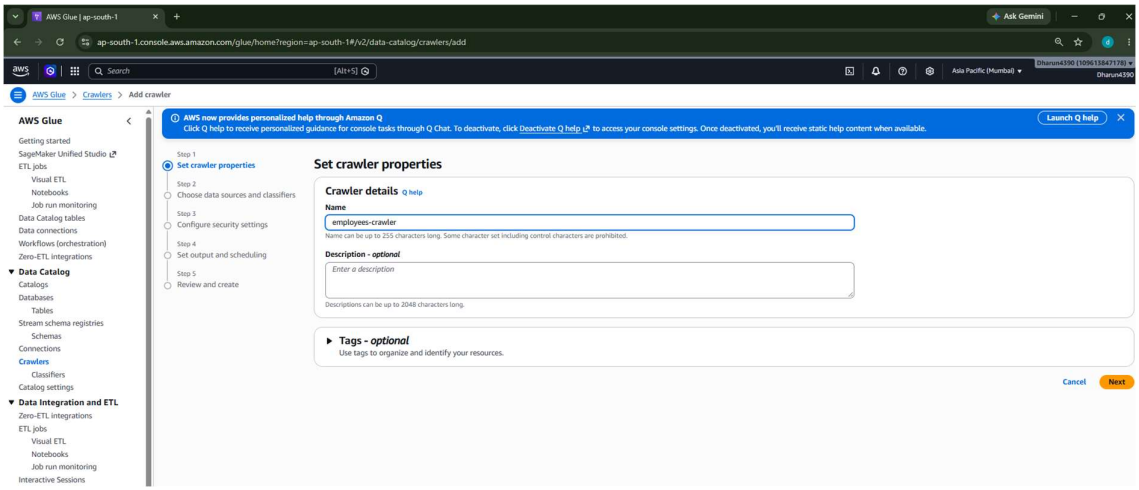


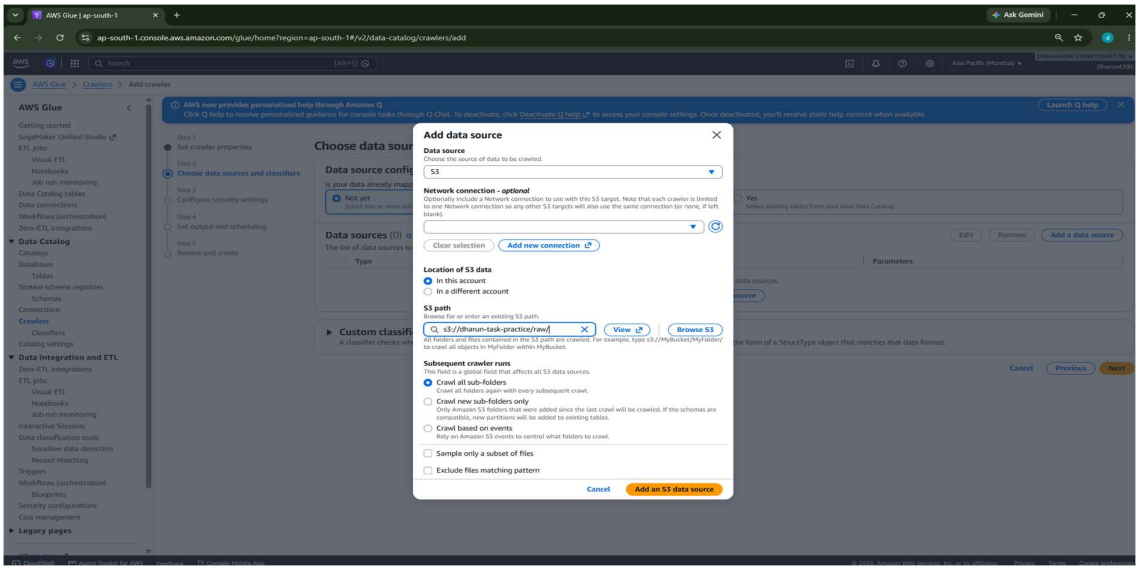


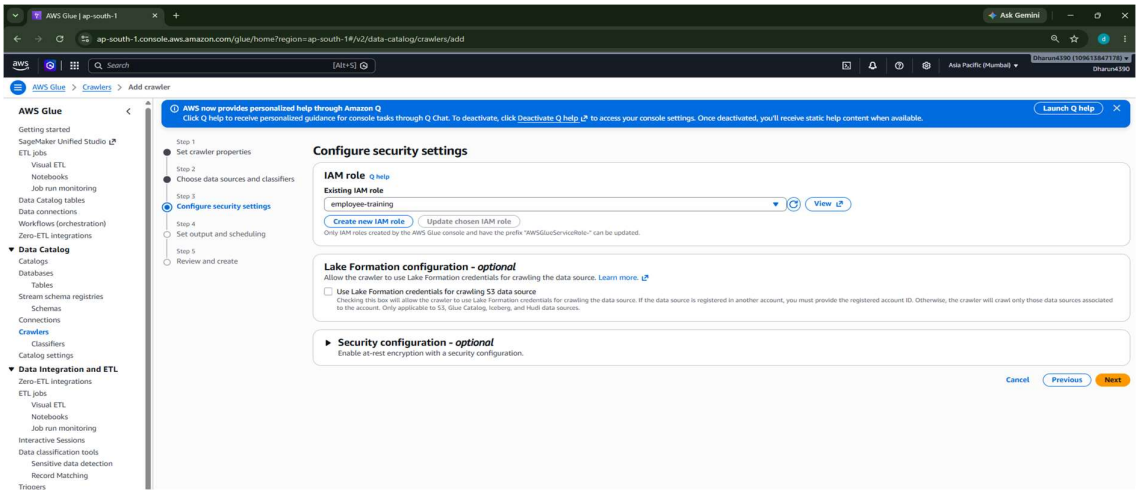
Glue Crawler

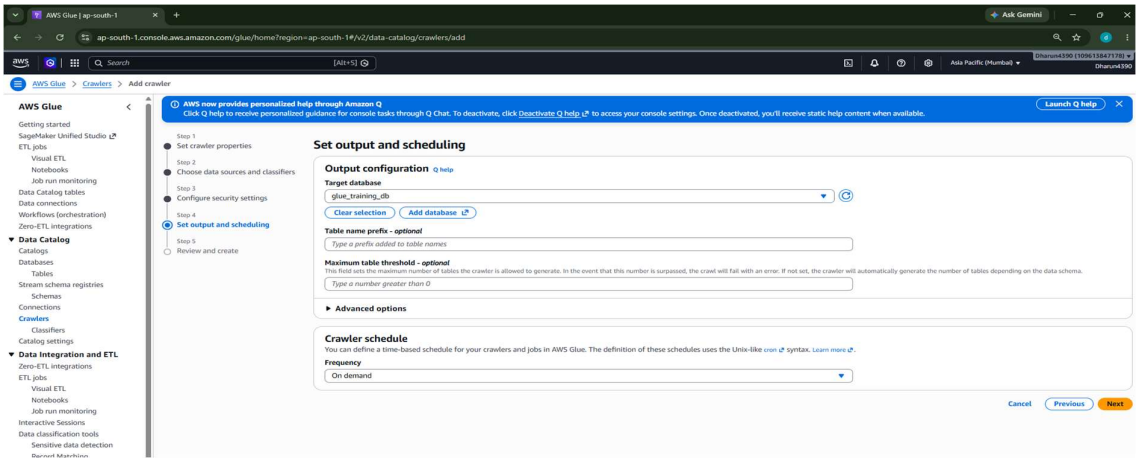



Crawlers - AWS Glue Console
ap-south-1.console.aws.amazon.com/glue/home?region=ap-south-1#/v2/data-catalog/crawlers
AWS Glue > Crawlers
AWS Glue
Crawlers
A crawler connects to a data store, progresses through a prioritized list of classifiers to determine the schema for your data, and then creates metadata tables in your data catalog.
Crawlers (0)
View and manage all available crawlers.
Create crawler
Launch Q help
No resources
No resources to display.

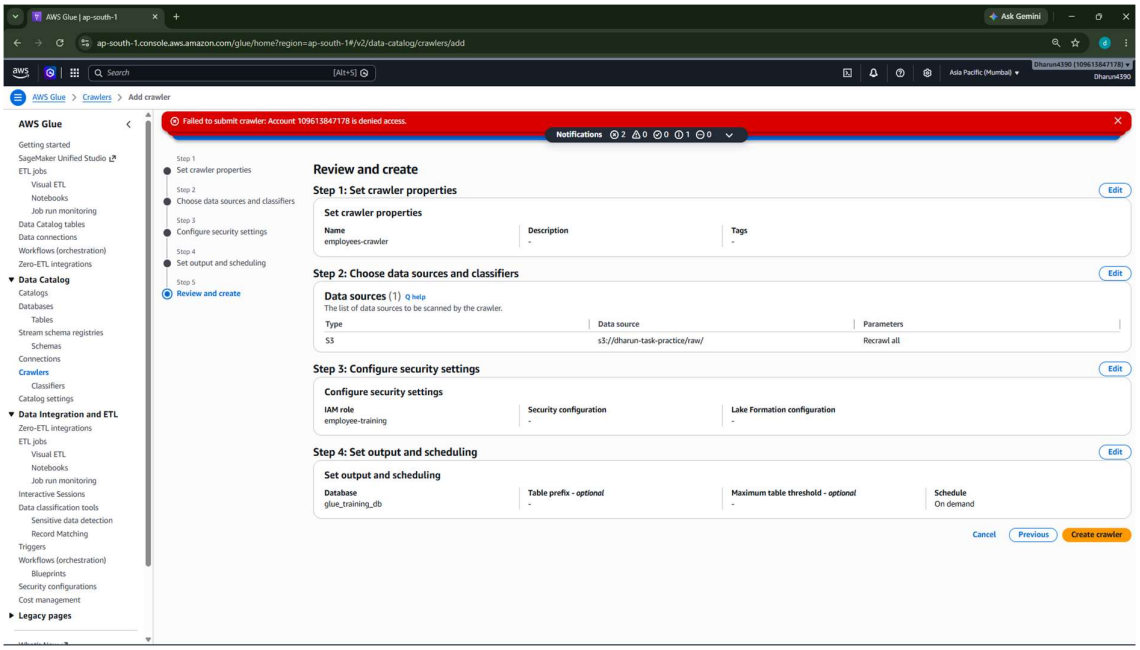

AWS Glue | ap-south-1
ap-south-1.console.aws.amazon.com/glue/home?region=ap-south-1#/v2/data-catalog/crawlers/add
AWS Glue > Crawlers > Add crawler
Set crawler properties
Crawler details
Name
employees-crawler
Description - optional
Tags - optional
Cancel
Next


AWS Glue | ap-south-1
ap-south-1.console.aws.amazon.com/glue/home?region=ap-south-1#/v2/data-catalog/crawlers/add
Add data source
Data source
S3
Network connection - optional
Location of S3 data
In this account
In a different account
S3 path
s3://dharun-task-practice/raw/
Browse S3
Subsequent crawler runs
Crawl all sub-folders
Crawl new sub-folders only
Crawl based on events
Sample only a subset of files
Exclude files matching pattern
Cancel
Add an S3 data source





Crawler Execution



Glue Data Catalog

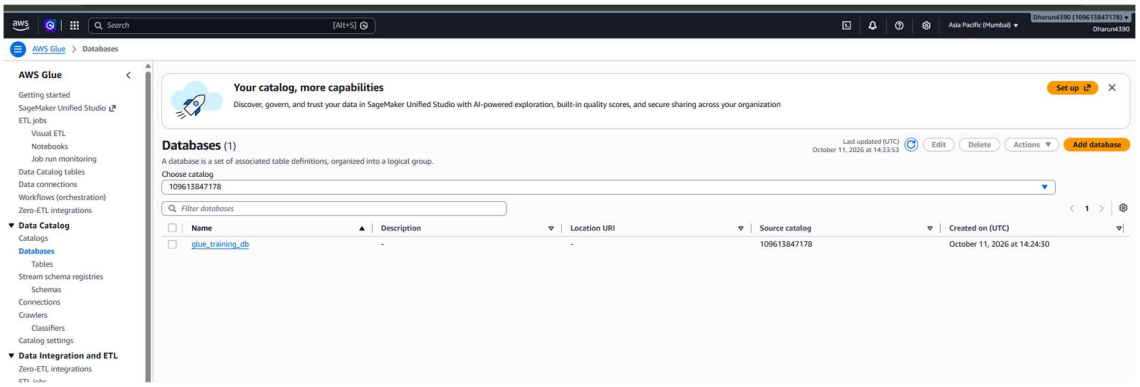
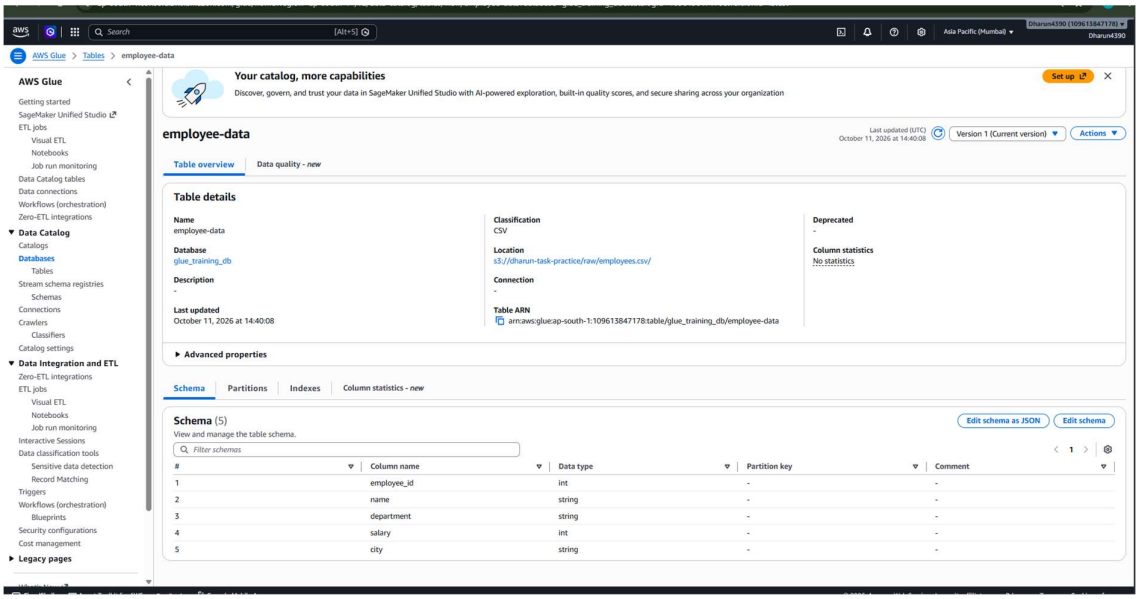
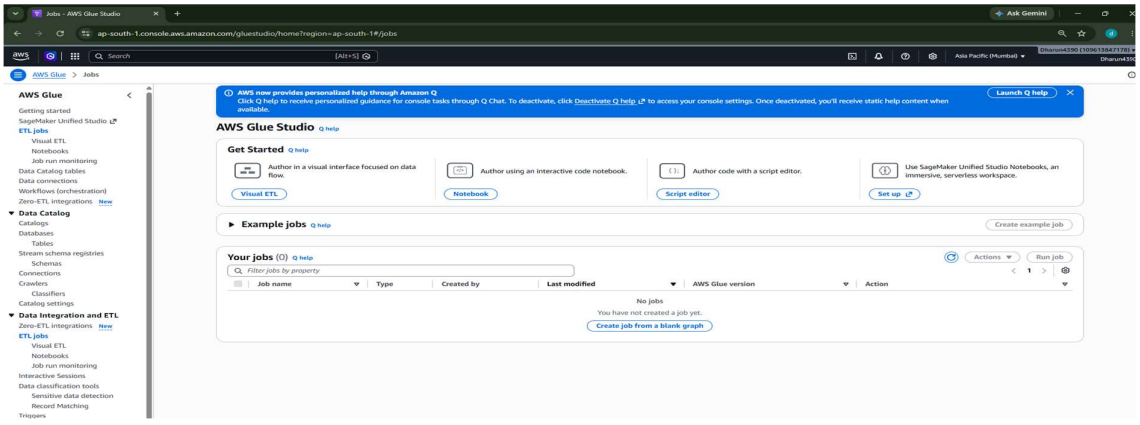
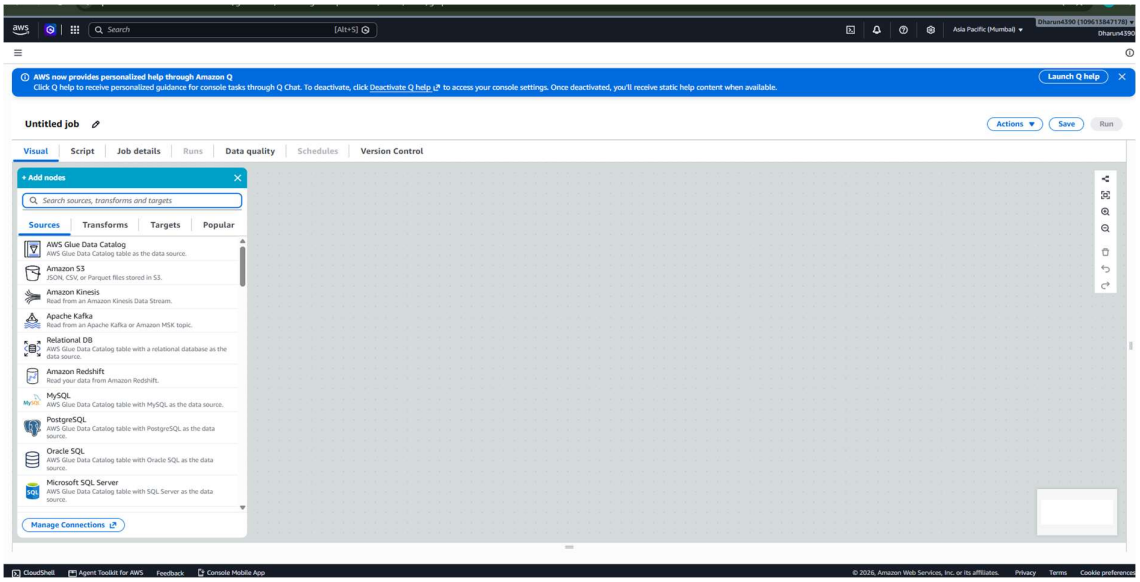


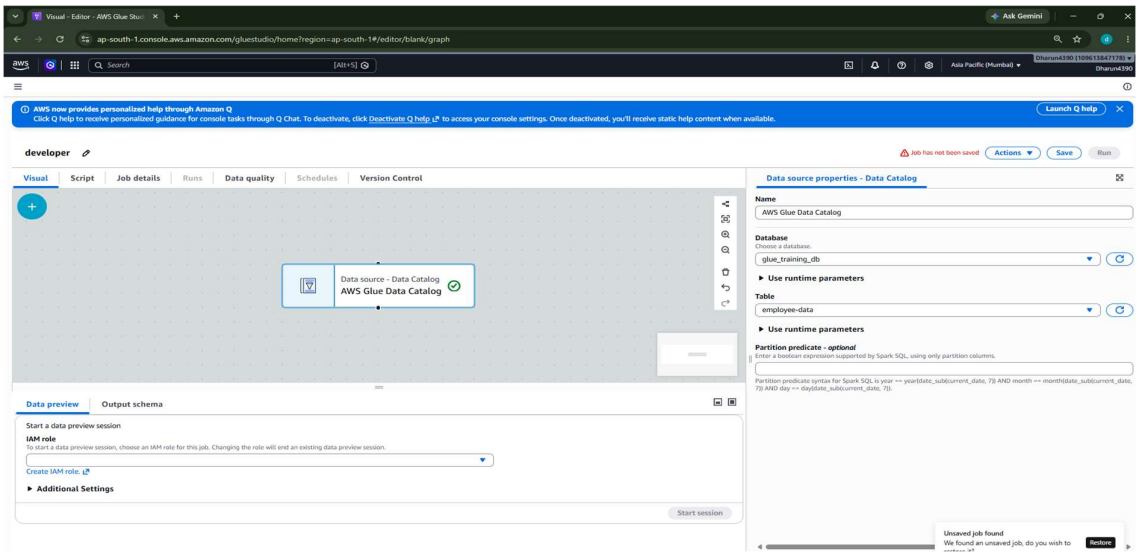
Table Schema

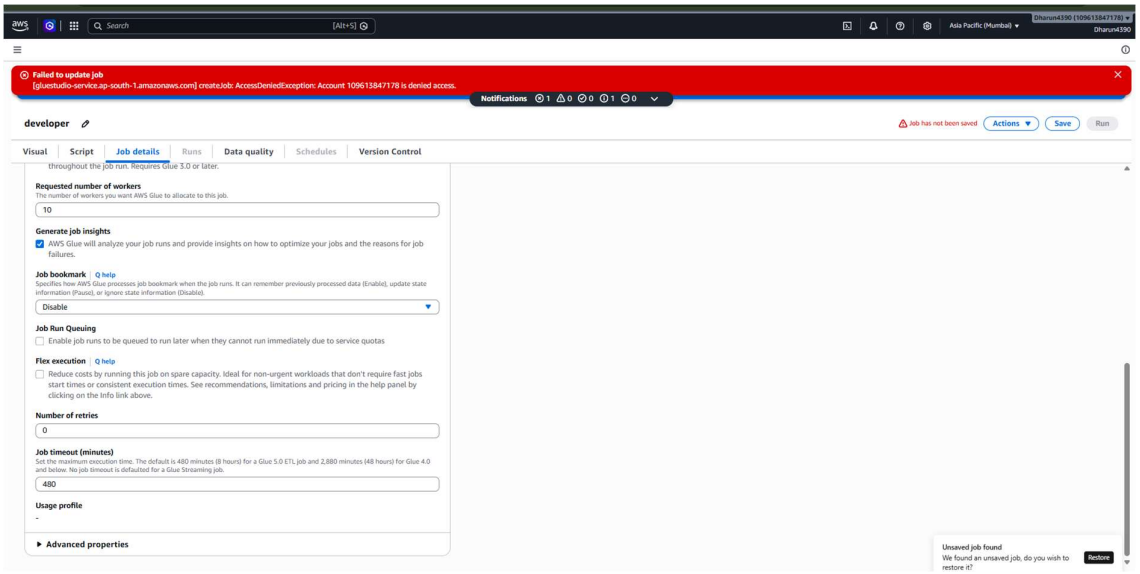


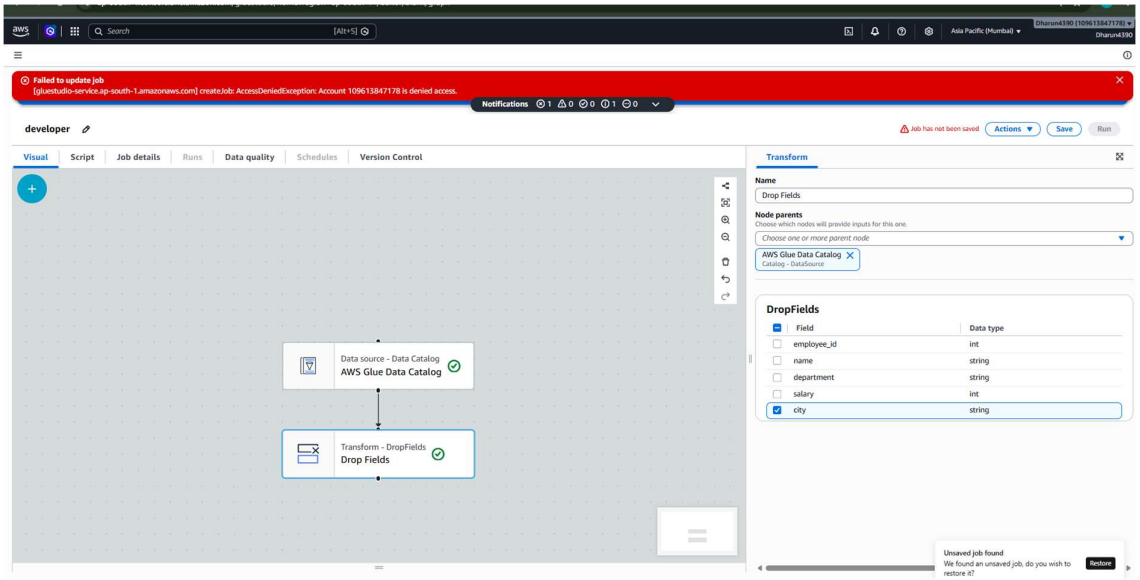
Glue Visual ETL

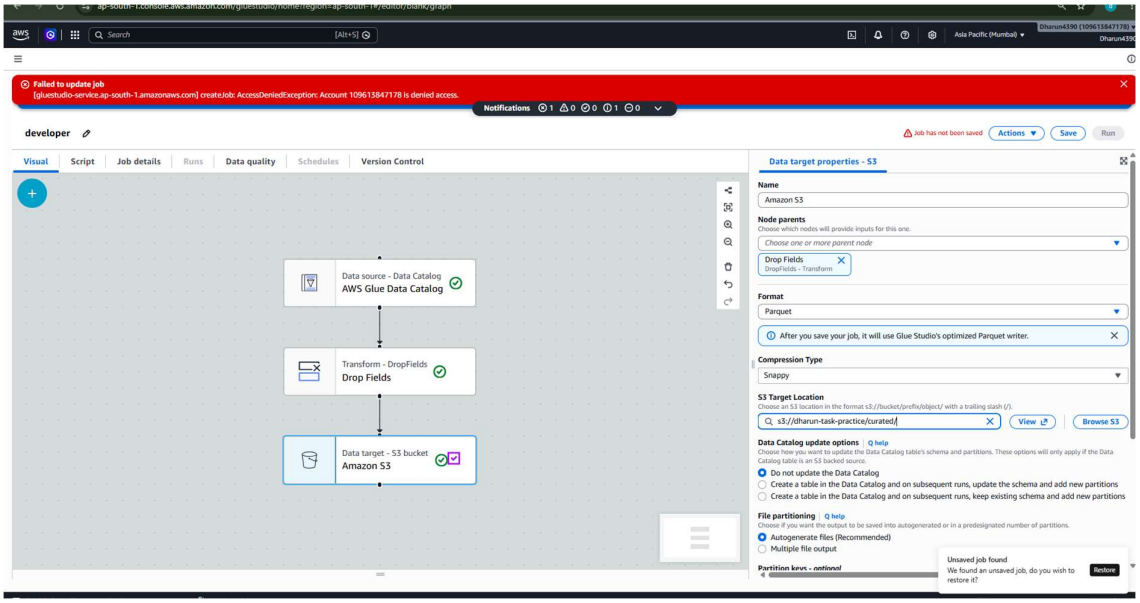



AWS now provides personalized help through Amazon Q
Click Q help to receive personalized guidance for console tasks through Q Chat. To deactivate, click Deactivate Q help to access your console settings. Once deactivated, you'll receive static help content when available.
Launch Q help
Untitled job
Actions
Save
Run
Visual
Script
Job details
Runs
Data quality
Schedules
Version Control
+ Add nodes
Search sources, transforms and targets
Sources
Transforms
Targets
Popular
AWS Glue Data Catalog
AWS Glue Data Catalog table as the data source.
Amazon S3
JSON, CSV, or Parquet files stored in S3.
Amazon Kinesis
Read from an Amazon Kinesis Data Stream.
Apache Kafka
Read from an Apache Kafka or Amazon MSK topic.
Relational DB
AWS Glue Data Catalog table with a relational database as the data source.
Amazon Redshift
Read your data from Amazon Redshift.
MySQL
AWS Glue Data Catalog table with MySQL as the data source.
PostgreSQL
AWS Glue Data Catalog table with PostgreSQL as the data source.
Oracle SQL
AWS Glue Data Catalog table with Oracle SQL as the data source.
Microsoft SQL Server
AWS Glue Data Catalog table with SQL Server as the data source.
Manage Connections
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Visual - Editor - AWS Glue Stu
Ask Gemini
ap-south-1.console.aws.amazon.com/gluestudio/home?region=ap-south-1#/editor/blank/graph
AWS now provides personalized help through Amazon Q
Click Q help to receive personalized guidance for console tasks through Q Chat. To deactivate, click Deactivate Q help to access your console settings. Once deactivated, you'll receive static help content when available.
Launch Q help
developer
Job has not been saved
Actions
Save
Run
Visual
Script
Job details
Runs
Data quality
Schedules
Version Control
Data source - Data Catalog
AWS Glue Data Catalog
Data source properties - Data Catalog
Name
AWS Glue Data Catalog
Database
Choose a database.
glue_training_db
Use runtime parameters
Table
employee-data
Use runtime parameters
Partition predicate - optional
Enter a boolean expression supported by Spark SQL, using only partition columns.
Partition predicate syntax for Spark SQL is year == year(date_sub(current_date, 7)) AND month == month(date_sub(current_date, 7)) AND day == day(date_sub(current_date, 7)).
Data preview
Output schema
Start a data preview session
IAM role
To start a data preview session, choose an IAM role for this job. Changing the role will end an existing data preview session.
Create IAM role.
Additional Settings
Start session
Unsaved job found
We found an unsaved job, do you wish to
Restore


Failed to update job
[gluestudio-service.ap-south-1.amazonaws.com] createJob: AccessDeniedException: Account 109613847178 is denied access.
Notifications
developer
Job has not been saved
Actions
Save
Run
Visual
Script
Job details
Runs
Data quality
Schedules
Version Control
throughout the job run. Requires Glue 3.0 or later.
Requested number of workers
The number of workers you want AWS Glue to allocate to this job.
10
Generate job insights
AWS Glue will analyze your job runs and provide insights on how to optimize your jobs and the reasons for job failures.
Job bookmark
Q help
Specifies how AWS Glue processes job bookmark when the job runs. It can remember previously processed data (Enable), update state information (Pause), or ignore state information (Disable).
Disable
Job Run Queuing
Enable job runs to be queued to run later when they cannot run immediately due to service quotas
Flex execution
Q help
Reduce costs by running this job on spare capacity. Ideal for non-urgent workloads that don't require fast jobs start times or consistent execution times. See recommendations, limitations and pricing in the help panel by clicking on the Info link above.
Number of retries
0
Job timeout (minutes)
Set the maximum execution time. The default is 480 minutes (8 hours) for a Glue 5.0 ETL job and 2,880 minutes (48 hours) for Glue 4.0 and below. No job timeout is defaulted for a Glue Streaming job.
480
Usage profile
-
Advanced properties
Unsaved job found
We found an unsaved job, do you wish to restore it?
Restore

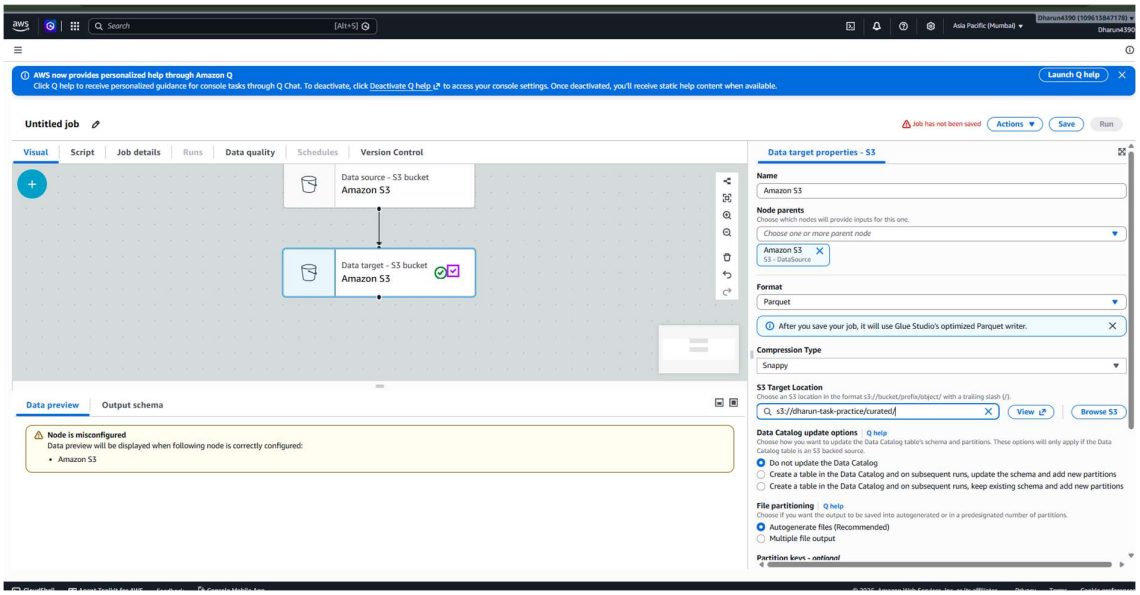




Glue Transformation

Since it is an free account ETL jobs cannot be created and due to it the Glue Transformation also cannot be done.

S3 Target Configuration



Glue ETL Job Execution

ETL job cannot be created as it is a free account.